

18

Electronic and Ionic Conductivities of Microtubules and Actin Filaments, Their Consequences for Cell Signaling and Applications to Bioelectronics

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18.1	Introduction	18-2
18.2	The Neuron.....	18-2
18.3	The Cytoskeleton	18-5
18.4	Overview of Biological Conductivity	18-7
18.5	Intrinsic Electronic Conductivity of Microtubules	18-9
18.6	Actin Filaments Support Nonlinear Ionic Waves	18-26
18.7	Long-Range Spatio-Temporal Ionic Waves Along Microtubules	18-30
18.8	Dendritic Cytoskeleton Information Processing Model	18-33
18.9	The Inter-Relation Between the Neural Cytoskeleton and the Membrane	18-35
18.10	Relationship to Cognitive Functions	18-36
18.11	The Potential for Bioelectronic Applications and Neuromorphic Computing.....	18-37
18.12	Discussion	18-39
	Acknowledgments	18-40
	References	18-40

18.1 Introduction

The prevailing view of neural information processing is based on passive properties of the membrane derived from the application of linear cable theory to dendrites, e.g., [114]. Recent studies, however, have suggested nonlinear models that accommodate new experimental evidence that appears to be inconsistent with this classical view. In particular, conventional theories do not satisfactorily explain: (a) significant fluctuations of synaptic efficacy over short periods of time in response to a recent burst of activation, e.g., [54], (b) a variety of active ion-channels capable of affecting the local membrane electric properties, and (c) nonlinear responses localized to specific dendritic sites, pointing to highly specialized mechanisms correlated with specific inputs. On the other hand, nonlinearities inherent in the new models, e.g., [74,133], give rise to a wider repertoire of (computational) capabilities, such as multiplication, fast correlation, etc. Moreover, experimental and theoretical support for conductive properties of protein filaments and the ionic clouds around them is becoming available, which indicates a far more complex picture of the signaling and information processing phenomena in neurons and other cells.

In this chapter, we suggest that functional electrodynamic interactions between cytoskeletal structures and ion-channels are central to the neural information processing mechanism. These interactions are supported by long-range ionic wave propagation along microtubule networks and actin filaments, and exhibit subcellular control of ionic channel activity, hence impacting the computational capabilities of the whole neural function. Cytoskeletal biopolymers, including actin filaments (AFs) and microtubules (MTs), constitute the backbone of wave propagation, and in turn interact with membrane components to modulate synaptic connections and membrane channels. Indeed, only recently clear functional interactions between these cytoskeletal structures have become apparent. Association of MTs with AFs in neuronal filopodia appears to guide microtubule growth, and plays a key role in neurite initiation [28]. This is further evidenced by the presence in neurons of proteins that interact with both MTs and F-actin, and proteins that can mediate signaling between both types of filaments. This is likely used to control microtubular invasion. The microtubule associated proteins MAP1B and MAP2, for example, are known to interact with actin in vitro, e.g., [146]. Cross-linking, MAP2 and/or MAP1B is highly probable to associate with both types of filaments contributing to the guidance of MTs along AF bundles. Extensive evidence exists confirming a direct interaction between AFs and ion channels, and a regulatory functional role associated with actin. Thus it is clear that the cytoskeleton has a direct connection to membrane components, in particular channels and synapses.

In the following, we demonstrate how each individual cytoskeletal component is capable of supporting both ionic and electronic wave propagation. In particular, we present a molecular dynamics model of the dendritic microtubule network (MTN) where arrays of MTs are interconnected by MAP2s. In this model, ionic waves propagate along the MTN and interact with C-termini of MTs to generate collective modes of behavior. Also in this chapter, a biophysical model of nonlinear ionic wave propagation along AFs is presented, supported by experimental evidence. Finally, we describe a model for a direct regulation of ion-channels and synaptic strength by AFs and networks of MTs that control the electrical response of the dendrite in particular, and the neuron in general. We begin this chapter by providing a general overview of the structure and function of a nerve cell, emphasizing the role of cytoskeletal filaments, ion channels, and their interactions.

18.2 The Neuron

The neuron is the quintessential communicating cell. A great variety of neural subtypes can be found interconnected to one another in complex electrochemical circuits via multiple synaptic connections. Thus, the amount of information processing by a neural circuit depends on the number of such connections or synapses. Neurons are highly polarized cells, whose level of morphological complexity increases during maturation into distinct subcellular domains. A typical neuron possesses three functional domains. These are the cell body or *soma* containing the nucleus and all major cytoplasmic organelles, a single axon, which extends away from the soma and takes cable like properties, and a variable number of dendrites, different

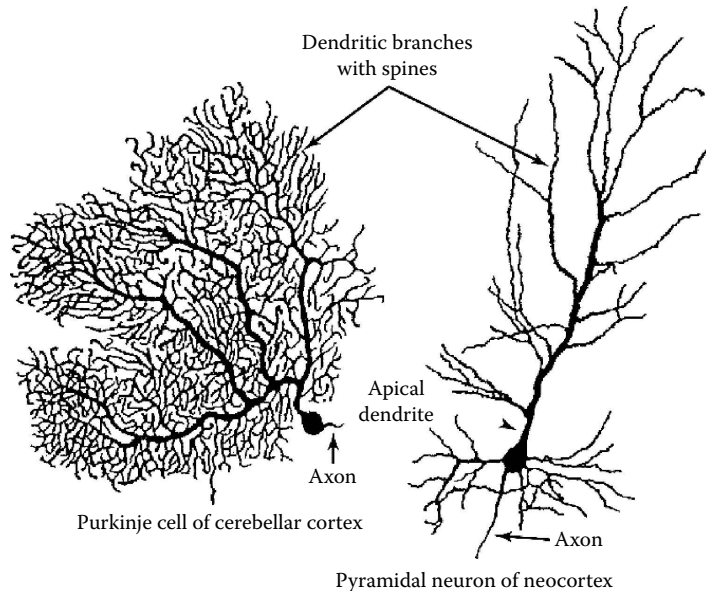


FIGURE 18.1 Two prominent examples of nerve cells.

in shape and complexity, which emanate from the soma. The axonal terminal region, where the synapse contacts it with other cells, displays a wide range of morphological specialization (see Figure 18.1).

Two major types of synapses can be found: (a) asymmetric, responsible for the transmission of excitatory impulses, and (b) symmetrical, present in inhibitory synapses. Turning morphology into a functional feature, the soma and dendritic tree are the major domains of receptive inputs. Thus, both the dendritic arborization and the axonal ramifications confer a high level of sub-cellular specificity in the localization of particular synaptic contacts on a given neuron. The extent to which a neuron may be interconnected largely depends on the three-dimensional spreading of the dendritic tree. Dendrites contain a large number of dendritic spines, which are small sac-like organelles, projecting from the dendritic trunk. Dendritic spines are more abundant in highly arborized cells such as pyramidal neurons, and scarcer in lowly interconnected smaller-sized dendritic tree inter-neurons.

Spines are the dendritic regions where most excitatory inputs are found, such that each spine contains one asymmetric synapse. The number of excitatory inputs can be correlated with the number of spines present in the dendrite. Dendritic spines are complex saccular enlargements, containing ribosomes, and cytoskeletal structures, including actin, and α - and β -tubulin. Electrically speaking, a single neuronal circuit is closed such that afferent inputs reaching the soma and the dendritic tree are transferred to the axon at the other pole of the cell which, in turn, is responsible for transmitting efferent neural information. Both the morphology of the axon and its course through the nervous system are correlated with the type of information processed by a particular neuron, and by its interconnectivity with other neurons. While the axon leaves the soma with a small swelling known as the axon hillock, some neurons protrude the axonal projection from the main dendrite instead. Cytoskeletal components are highly relevant in the structure/function of the neuron. The axonal hillock contains large “parallel” bundles of MTs. Axons are also located with other cytoskeletal structures, such as neurofilaments, which are more abundant in axons than in dendrites. The dynamics in cytoskeletal structures is central to our contention that information can be processed and “delivered” to the synaptic function by changes in the cytoskeletal structures.

Dendrites are the principal element responsible for synaptic integration and for the changes in synaptic strengths that take place as a function of neuronal activity. While dendrites are the principal sites for excitatory synaptic input, little is still known about their function. The activity patterns inherent to

the dendritic tree-like structures, such as in integration of synaptic inputs, are likely based in the wide diversity of shapes and sizes of the dendritic arborizations [116]. Some dendrites are unipolar, while others are multipolar; some have many orders of branching, while others have only one or two; and some branch primarily in two dimensions, while others have complex three-dimensional structures. The size and complexity of dendritic trees increase during development, which has been associated with the ability of the neural system to organize and process information [49]. The majority of synapses, both excitatory and inhibitory, terminate on dendrites, so it has long been assumed that dendrites somehow integrate the numerous inputs to produce single electrical outputs. It is increasingly clear that the morphological functional properties of dendrites are central to its integrative function. Branching in this context is essential to the increasingly complex ability of the neuron to respond to developmental complexity. Nerve branching and complexity are controlled by both external and internal cues. Of current interest to this are substrate and electromagnetic fields, while also recognized are the chemical gradients involved in local adaptation. Highly relevant to adaptation is cytoskeletal dynamics, as the dendritic cytoskeleton plays a central role in the process of ramification, filopodia formation, and more specialized neuronal activity such as long-term potential (LTP) and long-term depression (LTD) circuit dynamics, including memory formation.

The electrotonic or passive properties of dendrites became of great importance to our understanding of dendritic integration of synaptic signals. Early hypotheses were supported by work on motor neurons [34] and sensory receptors [38,67]. In that action, potentials were initiated in the axon hillock region as a result of the graded or algebraic summation of EPSPs and IPSPs occurring in various parts of the neuron. Because of the relatively short space constant calculated for motor neurons [18], it was predicted that excitatory post-synaptic potentials from synapses on the distal portions of dendrites would be ineffective in firing a neuron. Rall [115] revolutionized the field of dendrite physiology by using cable theory for the study of neurons and their dendrites. Action potentials initiated in dendrites by synaptic stimulation were later observed propagating in both directions at velocities of around 0.3 m/s. [7,21,43]. Direct intracellular recordings from dendrites of cerebellar Purkinje cells [84], cortical neurons [57], and hippocampal neurons [153] confirmed that action potentials could be initiated and propagated in dendrites. Very little is known, however, about the active, or voltage-dependent, properties of dendrites, and many questions remain unanswered. Such as the types of ion channels present in dendrites, location, and how active properties alter synaptic integration.

The firing of action potentials elicits characteristic M-shaped Ca^{2+} profiles across the neuron. The Ca^{2+} rise is small in the soma, highest in the proximal apical and basal dendrites, and small again in the distal apical dendrites [62,100,118–120]. The spatial distribution of Ca^{2+} channels explains the distinct Ca^{2+} rise in various cell locations but rather reflects the propagation into the dendrites of trains of action potentials. Imaging with Na^+ -sensitive dyes has yielded similar patterns, suggesting that Ca^{2+} influx into the dendrites is driven by Na^+ influx into the dendrites which is driven by Na^+ -dependent action potentials [62]. Ca^{2+} -dependent action potentials have been demonstrated in hippocampal neurons and dendrites [131]. Most of the action potential induced Ca^{2+} entry is mediated by voltage-gated Ca^{2+} channels [100]. Multiple types of voltage-gated Ca^{2+} channels have been demonstrated in hippocampal neurons [36,40,68,101]. Based on the addition of specific Ca^{2+} channel blockers, the presence of L-, N-, P-, Q-, R-, and T-type Ca^{2+} channels has been reported [16]. This channel distribution may contribute to spike-triggered Ca^{2+} entry into the soma and apical dendrites. Both low-voltage activated (LVA) channels and at least one high-voltage activated (HVA) channel seem to be most predominant in regions of the dendrite at least a 100 nm from the soma. Experimental results strongly suggest a heterogeneous distribution of different types of voltage-gated Ca^{2+} channels within the soma and dendrites of hippocampal neurons [152]. The presence of Na^+ channels in distal dendrites helps explain the back-propagation of action potentials well into the dendritic tree [88,89,138,139,141]. Action potential propagation in dendrites is rather low, and susceptible to the effects of cable filtering.

A most prominent feature of dendritic ion channel regulation is the potential boosting and enhancement of distal synaptic events. Voltage-gated Na^+ and Ca^{2+} channels may play an important role in amplifying the magnitude of EPSPs towards the soma. However, channel regulation may also be important for

dendritic interactions in the immediate vicinity of the synaptic input. Voltage-gated channels may alter the local input resistance and time constant, which in turn would influence both spatial and temporal summation of EPSPs and IPSPs, thus conveying highly nonlinear interactions; signal amplification [64,137]. Thus, the nonlinear properties of dendrites may serve as electronic devices. In addition to providing an output, the action potentials elicited by synaptic input, whether initiated at the local synaptic sites or at the axon hillock and back-propagated into the dendrites, would essentially reset the dendritic membrane potential. Action potentials might also provide feedback or an associative signal to other synapses active just before or just after the action potential.

Back-propagating action potentials may also provide feedback to dendritic synapses after a neuronal output has occurred. Ca^{2+} signals are implicated in post-synaptically-induced synaptic plasticity such as short-term potentiation (STP) [92], LTP, and LTD [8,15]. Actin has long been postulated to play important roles in neuronal morphogenesis. High concentrations of actin have been found in the leading edge of the growth cones of developing neurons, and in synapses of mature neurons. Actin filament dynamics also plays a role in a number of related phenomena, including axon initiation, growth, guidance, and branching, in the morphogenesis and stability of dendrites and dendritic spines in synapse formation and stability. Neurons conduct two types of cytoplasmic growth processes in axons and dendrites, both of which are functionally and morphologically distinct, but are developmentally led from the growth cone. Growth cones are composed of finger-like filopodia and veil-like lamellipodia that help the growing processes explore their environment. Axons travel long distances, and upon reaching their targets, produce terminal branches where the growth cones are converted into pre-synaptic terminals. Dendrites usually do not extend such long distances, but often branch extensively, giving rise to dendritic trees. Upon proper contact with axons, post-synaptic dendritic specializations create functional synapses by opposition with the pre-synaptic terminals of axons. These are localized to small protrusions on their dendritic trees called dendritic spines, which are the site of most excitatory synapses. In the mammalian brain these multiple synaptic connections are believed to play important roles in learning and memory. Thus, neural morphogenesis requires changes in both the underlying cytoskeleton and specific interactions with the plasma membrane. To generate a membrane protrusive structure such as an axon, coordinated changes in actin and microtubular organization are required. Growth cones of developing neurons, for example, rapidly extend and retract filopodia by concentrated changing in cytoskeletal dynamics, namely bundled AFs in filopodia, and composed of a cross-linked actin network in lamellipodia [80,81,155]. In this section, we have outlined the main features of signaling taking place within a neuron. Major emphasis was placed on the role played by the cytoskeleton in the neuron's structure, growth, and internal signaling. The next section discusses the main biophysical properties of the cytoskeleton.

18.3 The Cytoskeleton

The cytoskeleton is a major component of all living cells. It is made up of three different types of filamental structures, including actin-based microfilaments (MFs), intermediate filaments (IFs) (e.g., neurofilaments, keratin), and tubulin-based microtubules (MTs). All of them are organized into networks, which are interconnected through numerous particular proteins, and which have specific roles to play in the functioning of the cell. The cytoskeletal networks are mainly involved in the organization of different directed movements in cell migration, cell division, or in the internal transport of materials. Polymerization of MFs is responsible for cell migration and for the remodeling of the leading edge of cells. Molecular motors are protein complexes associated with the cytoskeleton and drive organelles along MTs and MFs in a "vectorial" transport.

MTs are some of the most basic and important cytoskeletal elements in the morphological scheme of neurons. As for the case of cytochalasins in actin structures, the addition of microtubular modifying agents was first used to assess the role of MTs in axon elongation. Microtubular disrupting agents affect axons first and have no early effects on growth cones and filopodia, suggesting that MTs provide support for the axon. Conversely, microtubular invasion may be a critical element in the formation of neurite projections. Initially, expanded lamellipodia first undergo segmentation at certain spots where MTs accumulate in an

ordered array [27,144,156] and gradually “migrate away” from the cell body. These lamellipodia transform into nascent growth cones. Neurite elongation takes place as the MTs in the shaft become compressed into a narrower bundle. In cultured hippocampal and cortical neurons, this process most likely implies the coordinated control of the major cytoskeletal components, including actin, MTs, and associated proteins.

Thus, MTs are essential for neurite development. One particular aspect of microtubular involvement in neurite outgrowth is vesicular transport. Most membrane insertion occurs at the growth cone of axons [22,24]. MAP-stabilized MTs, form parallel or antiparallel arrays, which might act as compression resistant struts inside neurites [61]. Because MTs constantly invade the actin cytoskeleton in lamellipodia of epithelial [125] and neuroblastoma cells [27], such a mechanical role for MTs could also be important during de novo neurite initiation. Neurite-like protrusions, for example, can be induced in non-neuronal cells by the exogenous expression of stabilizing proteins such as MAP2 and tau [72,79,157]. Spontaneous neurite initiation is dependent on both the presence and the dynamic properties of MTs.

A number of proteins associate with MTs, including MT motors, kinases, so-called “structural” MAPs, which alter microtubule structure, and specialized MAPs, which bind to microtubule plus or minus ends. Members of the MAP2/tau family have long been proposed to be important regulators of neurite behavior [10,48,151]. MAP2 and tau stabilize MTs by reducing catastrophe and promoting rescues, leading to prolonged growth periods and thus enhanced net microtubule accumulation [32,45,75]. MAP2 and tau stabilize MTs by binding along the outer ridges of protofilaments [2].

F-actin and MTs cytoskeletal networks are typically thought to fulfill separate, independent cellular roles. Highly dynamic actin networks are known for their role in cell spreading and contraction. The more stable MT cytoskeleton is known for its importance in cell division and organelle trafficking. However, recent studies provide important roles for the actin cytoskeleton in cell division and trafficking and for MTs in the generation and plasticity of cellular morphology. A direct physical association between both cytoskeletons has been suggested, because MTs often preferentially grow along actin bundles and transiently target actin-rich adhesion complexes.

Association of MTs with actin cables in filopodia appears to guide microtubule growth along the most efficient path anti-parallel to retrograde flow [130]. Thus, specific coupling between MTs and actin bundles presumably promotes MT advance. Proteins that are able to interact with both MTs and F-actin, or proteins that can mediate signaling between both cytoskeletons likely control the regulation of MT invasion. This is further strengthened by specific microtubule association of the microtubule-associated proteins MAP1B and MAP2, both known to interact with actin in vitro [20,109,128,134,146]. Furthermore, MAP2 binds to F-actin and efficiently induces actin bundle formation [51,129]. It is likely that by crosslinking, MAP2 and/or MAP1B associated with both cytoskeletons could be involved in the guidance of MTs along actin filament bundles. Alternatively, MAPs could shuttle from MTs to actin and could alter F-actin behavior by actively crosslinking AFs.

The effect of changes in the cytoskeleton on ion channel activity has been the focus of recent attention. Most studies have employed drugs that selectively stabilize or destabilize either AFs or MTs, with resulting effects on specific ion channel activity, either as changes in whole cell conductance or single channel activity by the patch clamping techniques. Although direct binding of specific cytoskeletal proteins to individual channel proteins remains to be demonstrated, recent work in which exogenous purified actin has been added to the cytoplasmic side of channels excised from the cell membrane has in numerous cases directly confirmed an effect of cytoskeletal filaments. A body of earlier evidence suggested that various cytoskeletal components, including actin and actin-associated proteins, anchor, co-localize, and regulate both the spatial stability as well as the function of ion transport proteins. It is possible that actin either binds directly to the channels, or that actin may first interact with actin-binding proteins, which in turn regulate (by binding or other indirect interaction) ion channel function. Independent studies demonstrated that membrane-resident CFTR is functionally regulated by AF organization, which may also involve a direct interaction with actin. Thus, ion channel may be controlled by direct and indirect cytoskeletal interactions.

Before we present an integrated view of the conductive properties of the active and passive structural blocks of the cytoskeleton, we should review the current state of knowledge regarding their conductive properties.

18.4 Overview of Biological Conductivity

Cope [19] has presented a detailed review of the applications of solid-state physics concepts to biological systems. He divided the discussion into seven groups. The first involves semiconduction of electrons across enzyme particles as a rate-limiting process in cytochrome oxidase, evidenced by kinetic patterns of enzymes and microwave Hall measurements. Second, pn-junction conduction electrons were suggested by kinetics of photobiological free radicals in the eye and in photosynthesis. Their I-V characteristics conform to the diode equation:

$$I = I_0 \exp(V/kT) \quad (18.1)$$

Third, there may be a connection between superconduction and growth in nerves. Fourthly, phonons and polarons are involved in mitochondrial phosphorylation. Next, piezoelectricity and pyroelectricity may be involved in the growth of nerves and bone structures. Then, infrared electromagnetic waves may transmit energy in lipid bilayers of nerves and mitochondria. Finally, complex sodium and potassium ions in structured cell water may be analogous to valence band electrons in semiconductors, free cations being analogous to conduction electrons. The ionic process in cell water resembles electronic conduction in semiconductors. For dried proteins or DNA, Szent-Gyorgi [142] noticed that their conductivities obeyed the relationship:

$$\sigma = \sigma_0 \exp(-E_a/kT) \quad (18.2)$$

With activation energies E_a on the order of 1 eV. The presence of water may cause considerable changes in conductivities, (for example, in cytochrome oxidase this may be reduced to 0.3 eV).

Adessi *et al.* [1] have provided a theoretical treatment of transmission through a poly(g)DNA molecule. They found that there was a modification of the rise of a B-DNA form, which can induce a shift of the conduction channel towards a valence one. They found that deformation of the backbone of the molecule has a significance on the hole transport. Furthermore, the presence of ionic species (e.g., Na) in the surrounding of the molecule can create new conduction channels. In photo-excitation experiments, hole transport occurs through coherent transport at short distances and long-range incoherent or hopping transport at large distances. Fink *et al.* [39] studied the metallic behavior in λ DNA using an electron projection technique. Superconducting behavior has been observed by Kasumov *et al.* [66]. Porath *et al.* [111] observed semiconductor behavior with a poly(g)-poly(c)DNA molecule. A scanning force microscope has been used by de Pablo *et al.* [26] to observe insulating behavior. This paper models a single poly(g) DNA lying between electrodes, and transmission was computed with a Landauer-Büttiker formalism based on Green's functions. The electronic properties of O_2 -doped DNA have been studied by Mehrez *et al.* [96], who suggested that DNA conduction is affected by O_2 -doping, in particular hole doping. However, ab initio atomic structure calculations of O_2 -doped DNA indicate negligible charge transfer. The absence of dc conductivity in λ DNA has been investigated by de Pablo *et al.* [26]. First principles electronic structure calculations indicate a minimum DNA resistance of $10^{16} \Omega$ per molecule, and a minimum resistivity of $10^4 \Omega m$. However, lower energy electron bombardment may dramatically increase conductivity and with electron energies between 50 and 200 eV may lead to metallic conduction with a resistance of $2 \times 10^8 \Omega$. Fink *et al.* [39] discussing DNA and conducting electrons stated that I-V characteristics of DNA indicate ohmic-like behavior with a resistance of $2 M\Omega$ for a 600 nm-long DNA rope corresponding to $\rho = m\Omega \text{ cm}$. These authors speculate that DNA may be semiconducting, insulating, or metallic simultaneously depending on the arrangement of molecules in the structure. Endres *et al.* [37] discussed the quest for high-conductance DNA saying that DNA conduction is an unsolved problem first enunciated by Eley and Spivey [35]. Experimental outcomes range from insulating to semiconducting to even superconducting, and the uncertainties depend on: (a) contacts between the electrodes and the DNA and (b) differences in the molecules (sequence, length, arrangement, preparation) within the DNA. It appears that drying DNA can lead to conformations with localized electronic high states, although hole doping of the backbone by counter-ions might also be possible. Wet DNA may support electrical current due to solvent impurity

states in the large high- Π^* energy gap. Davies and Inglesfield [25] discussed the embedding method for conductance of DNA calculations. It is believed that the pathway for transport in DNA runs through the bases in the center of the double helix molecule. The dominant transport mechanism for DNA appears to be coherent hopping. Calculations show extreme sensitivity to the choice of contact orbitals.

Marechal [95] discussed the transfer of protons as a third fundamental property of hydrogen bonds. The importance of the cooperative resonance types in cyclic structures is stressed. Proton transfers along H-bonds have been discussed as a mechanism of vital importance for chains, helices, and in cyclic structures such as rings. Consta and Kapral [17] discussed proton transfer in mesoscopic molecular systems. Proton transfer rates and mechanics are studied theoretically with solvent molecules in solution. The proton transfer occurs as a result of orientational fluctuations on the cluster's surface. The environment influences solvent effects and reaction rates. The rate constant was established to be 1/59 ps at 260 K and for the bond length fixed at 2 Å. Morowitz [102] discussed proton semiconductors and energy transduction in biological systems. He presented the possibility of protochemistry paralleling electrochemistry. ATP synthesis was discussed as coupled to proton transport assuming a gated protein semiconductor across a membrane. Protein conduction in biological structures involves the water of hydration and protein molecules.

Ripoll *et al.* [121] investigated ionic condensation and signal transduction. Ion condensation on biopolymers at a critical value of the charge density may lead to intra-cellular signaling due to the diffusion of condensed counter-ions in the near region along cytoskeletal filaments. A feedback mechanism was proposed between condensation/decondensation of calcium and the activation of calcium-dependent enzymes. This is shown to create coherent patterns of protein phosphorylation. It was postulated that ion condensation operates in signal transduction. In the case of polymer ionics [117], the conductivity depends on temperature via the formula:

$$\sigma T = A \exp[-B/(T - T_0)] \quad (18.3)$$

indicating activated hopping. The concept of the Fermi level in solution has been extended by Bockris and Khan [9] through the introduction of a reversible potential for a redox couple.

Unfortunately, measurements of conductivity of biopolymers (including MTs) are fraught with difficulties due to their structural heterogeneity, strong dependence on the environmental conditions (pH, ionic concentration, temperature, etc.), and the inherent liquid state of the sample. Nonetheless, attempts have been made recently to measure resistance values of MTs, both intrinsically and as ionic conduction cables. Fritzsche *et al.* [41] made electrical contacts to single MTs following dry-etching of a substrate containing gold microelectrodes. Their results indicate intrinsic resistance of a 12 micron-long MT to be in the range of 500 megaohms, giving a value of resistivity of approximately 40 mega-ohms/micron in their dry state. The same group performed measurements on gold-coated MTs [42] which were covered with a 30-nm layer of gold following sputtering. The resistance of the thus metallized MTs was estimated to be below 50 ohms, (i.e., originated entirely due to the metallic coating). Minoura and Muto [99] measured the ionic conductivity of MTs using an electro-orientation method under an alternating electric field. The ionic MT conductivity was calculated as 1.5 ± 0.5 mS/m, which is approximately 15 times greater than that of the buffer solution. This was attributed to the counter-ion polarization and is consistent with another recent study of ionic conductivity along MTs. Furthermore, Goddard and Whittier [47] reported their measurements of RF reflectance spectroscopy of samples containing the buffer solution, free tubulin in buffer, MTs in buffer, and finally, MTs with MAPs in buffer. The sample size in each case was 5 ml. The concentration of tubulin was 5 mg/ml and in the last case the concentration of MAP2 and the tau protein was 0.3 mg/ml. The average DC resistance reported by these authors was (a) 0.999 k Ω (buffer), (b) 0.424 k Ω (tubulin), (c) 0.883 k Ω (MTs), and (d) 0.836 k Ω (MTs + MAPs). It's not straightforward to translate these results into resistivity of MTs without making assumptions about their geometrical arrangement and connectivity. However, assuming all tubulin being polymerized in case (c) and a uniform distribution of MTs forming a combination of parallel and series networks, one can find the resistance of a 10 μ m-long MT, forming a basic electrical element in such a circuit, to have approximately an 8 M Ω resistance. This compares very favorably to an early theoretical estimate of MT conductivity based on the Hubbard model with electron hopping between tubulin monomers. This model predicted the resistance of a 1- μ m MT

to be in the range of $200\text{ k}\Omega$, hence a $10\text{-}\mu\text{m}$ MT would be expected to have an intrinsic resistance of $2\text{ M}\Omega$, within the same order of magnitude as the result reported by Goddard and Whittier [47]. While direct experimental analysis of intrinsic conductivity of the various cytoskeletal components is still nebulous, theoretical modeling of these effects is very well developed. The next section presents our current understanding of the intrinsic conductivity of MTs.

18.5 Intrinsic Electronic Conductivity of Microtubules

Microtubules (MTs) are ubiquitous in eukaryotic cell biology. Their primary role is to serve as the cellular scaffold and act as a corridor for the transport of vesicles through the cell by motor proteins. The interest in MTs stems from their involvement in a number of crucial cellular processes including mitosis and, more recently, the discovery of their role in the communication between the exterior of the cell and the nucleus [46,94]. The special arrangement of MTs just prior to chromosome separation makes them an ideal candidate for mediating a signal of some sort, which would cause the coordinated breakage of sister chromatids. In the axons of nerve cells, the MTs are arranged in bundles, parallel to the axon [56]. These groups of MTs span the entire length of the axon and might serve to carry signals from the cell body to the nerve terminus or vice versa. No mechanism for this sort of direct feedback is known. We demonstrate here that the MT may be able to conduct a current along its length through a process of electron hopping between binding sites on each dimer.

The MT is made up of a protein called tubulin. Each subunit contains α -tubulin and β -tubulin monomers joined together to form a dimer. The dimers are connected end to end to form a protofilament. These protofilaments, usually 13 in number, are wrapped up to form a hollow tube, the MT. The dimers are themselves 5 nm in diameter and have a length of about 8 nm . A MT has an outer diameter of 25 nm and an inner diameter of 15 nm (see Figure 18.2).

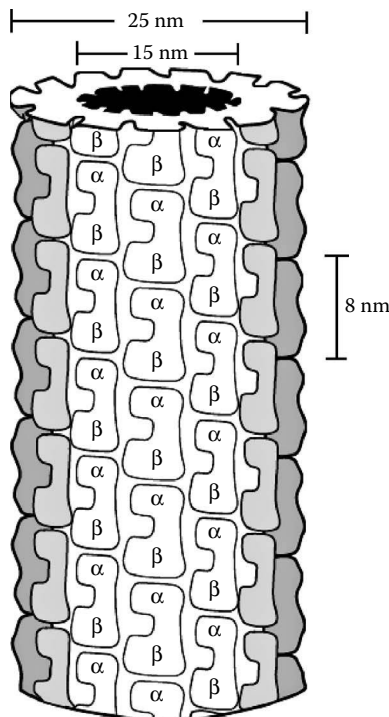


FIGURE 18.2 Structure of a typical microtubule demonstrating how it is formed by a special arrangement of tubulin dimers.

The tubulin dimer of which the MT is composed has at least two different conformational states [58]. That is, while the amino-acid chain of the protein remains fixed, its three-dimensional shape may change. It is believed that these conformational changes are the result of hydrolysis and an associated charge movement from one binding site within the molecule to another. It is precisely this charge movement that may lead to the conduction through the polymer.

One may take a semi-classical view of the movement of electrons along the protofilament. We ignore the fact that the protofilament is nearly one-dimensional and consider its *bulk* properties. Assuming two conduction electrons for each tubulin dimer, the conduction electron density is

$$\begin{aligned} n_0 &= 2/(8nm \times 5nm \times 5nm) \\ &= 1.0 \times 10^{19}/cm^3. \end{aligned} \quad (18.4)$$

In comparison with other materials, this value is slightly higher than typical semi-conductor concentrations, usually $10^{13} - 10^{17}/cm^3$, but is much less than typical metals, which have a conduction electron density of $10^{22} - 10^{23}/cm^3$. **However, the electrons are expected to have relatively low mobilities unless they are excited from the ground state configuration.** The excitation energy is a function of both the number of electrons within the protofilament and its size. Suppose that the mobility of excited electrons is at least similar to the mobility of electrons within semiconductors, such as Si where the electron mobility, μ_e , is about $1300 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. The electrical conductivity σ is given in a semi-classical model by the well-known formula

$$\sigma = ne\mu_e \quad (18.5)$$

where e is the electron charge and n is the conduction electron density given by

$$n = n_0 e^{-\Delta/\kappa T} \quad (18.6)$$

where n_0 represents the total number of electrons. If we assume that the gap between valence and conduction bands, Δ , is about 0.40 eV, then at physiological temperature, some of the carriers will be excited and should be available to conduct. Applying the preceding formula (18.6), a conductivity of about 10^{-2} (in SI units) is expected for the protofilament.

$$\sigma \sim (10^{25} e^{-0.40/0.026})(1.6 \times 10^{-19})(0.13) \sim 0.04 \Omega^{-1} m^{-1} \quad (18.7)$$

However, if the hybridization of electronic orbitals lowers the gap to a value closer to zero, the expected conductivity would be significantly higher:

$$\sigma \sim (10^{25})(1.6 \times 10^{-19})(0.13) \sim 2 \times 10^5 \Omega^{-1} m^{-1} \quad (18.8)$$

These values should be compared to values in excess of 10^8 for copper and other good conductors, and 10^7 for lesser conductors and semi-metals. Typical semi-conductors such as Si and Ge have intrinsic conductivities of $10^{-3} - 10^{+1}$ at room temperature, but when heavily doped, the conductivity may rise to between 10^4 and 10^5 .

In order to model the MT as a semiconductor, the individual tubulin dimer must first be physically analyzed. The $\alpha\beta$ -tubulin dimer is comprised of about 890 amino acids which represents about 13000 nuclei. A gross simplification which we introduce is to treat each monomer as having an effective site energy for electronic binding. Fluctuations of the site energy on scales less than the monomer spacing are therefore washed out. It is convenient to model this system as two quantum wells separated by a potential barrier. The two wells represent two binding sites for an electron, which may hop between the monomers, resulting in the change of molecular conformation. We can estimate the relevant parameters of the well depth and the well width from kinetic studies and with a knowledge of the molecule's overall geometry. Once the bound states and their energies are known for this system, this knowledge may be applied to a

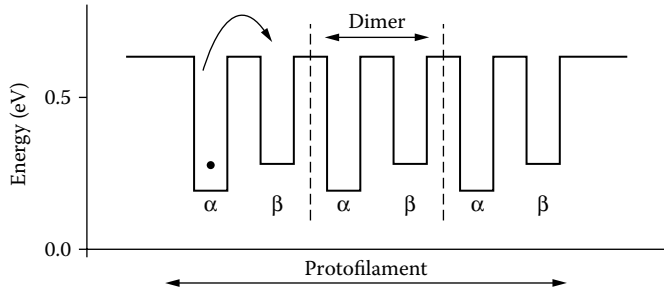


FIGURE 18.3 A one-dimensional chain of quantum wells representing the energy profile along a MT protofilament where the α and β monomers have alternating site energies.

second-quantization scheme. This procedure allows us to make progress without specifically requiring us to solve for the wavefunction of the system.

A 1D chain of quantum double wells is pictured in Figure 18.3. For simplicity, we are initially making each binding site have the same energy. If this is related to GTP hydrolysis, they may differ by about 0.17 eV (4 kcal/mol) [59]. This is roughly the quantity of energy stored in the lattice when GTP is hydrolyzed to GDP [13,60]. The barrier height between dimers has been estimated from kinetic data. The bond strength must be somewhat larger than the hydrolysis energy delivered to the lattice or else hydrolysis of the dimers would cause immediate disassembly. Hopping along the protofilament by electrons presumably has an energy comparable to this binding energy. We have estimated 0.4 eV for intra-dimer hopping and 1.0 eV for the inter-dimer hopping. What may be varied in the model, is the central barrier which must be overcome for each tubulin dimer to change conformation. If it is nearly as large as the barrier to electron movement between dimers, then the protofilament will look like a chain of single quantum wells rather than double quantum wells because the difference between the α and β monomers will not be distinguished. However, if the central barrier is too small, the chain will again look like a series of single quantum wells since the effect of the barrier would be negligible. The depth of the deeper well, f_1 , has been selected as 0.4 eV initially. Given the value, we assumed a 25% difference in well depths. If f_1 is much larger, the dimer nature of the protofilament is lost because the wells are effectively the same depth.

The results for the single electron eigenstates of a double quantum-well may be found using a transfer matrix method for the time-independent 1D Schrodinger equation in each interval in which the piece-wise constant potential is defined as

$$\hat{H}\psi = \frac{\hat{p}^2}{2m}\psi + \hat{V}\psi = E\psi \quad (18.9)$$

The oscillatory solution

$$\psi(x) = A \cos(kx) + B \sin(kx) \quad (18.10)$$

applies to $E > V$ where

$$k = a\sqrt{2m(E - V)}/\hbar \quad (18.11)$$

while for $E < V$, the solutions are exponentially damped

$$\psi(x) = Ae^{\kappa x} + Be^{-\kappa x} \quad (18.12)$$

where

$$k = \sqrt{2m(V - E)}/\hbar \quad (18.13)$$

Now, matching conditions for the wave function and its derivative are applied at each interface. This gives a 2×2 matrix relating the coefficients in the wave function expansion according to

$$\begin{pmatrix} A_2 \\ B_2 \end{pmatrix} = \begin{pmatrix} {}^{2,1}M_{11} & {}^{2,1}M_{12} \\ {}^{2,1}M_{21} & {}^{2,1}M_{22} \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \end{pmatrix} \quad (18.14)$$

In the preceding equation, the superscript to the left of the M denotes which coefficients are being linked. The subscripts indicate the matrix element. In an iterated fashion, one eventually arrives at the following:

$$\begin{pmatrix} A_n \\ B_n \end{pmatrix} = \begin{pmatrix} {}^{n,1}M_{11} & {}^{n,1}M_{12} \\ {}^{n,1}M_{21} & {}^{n,1}M_{22} \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \end{pmatrix} \quad (18.15)$$

Now, for bound states $A_1 = B_n = 0$ so that the wavefunction decays outside the region of the quantum wells:

$$B_n = 0 \quad \Rightarrow \quad {}^{n,1}M_{22} = 0 \quad (18.16)$$

Given the choices for our quantum double well (width, $a = 2$ nm; and height, $V = 1.0$ eV), one finds that there are two bound states for this dimer problem. As the number of dimers is increased, one resorts to numerical analysis. In Figure 18.4, the energy of the two bound states is plotted as the energy of the central barrier is varied. The lowest energy state is the symmetric wavefunction, which increases in energy sharply as the central barrier is increased. The second bound state is the antisymmetric wavefunction it has a node within the barrier, hence its energy increases only marginally.

The energy calculation for a sequence of n such dimers leads to the development of bands as hybrid orbitals develop across the polymer. Roughly speaking, one band is of hybridized symmetric orbitals and the higher lying band of hybridized antisymmetric orbitals. The separation of the two bands and the spread of each individual band determine the values of the parameters which shall be used in the tight-binding Hubbard model used to model the polymer.

The second-quantized Hamiltonian for a many-electron multidimer system becomes

$$\hat{H} = \sum_{i,\sigma} \epsilon_i \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} + \sum_{i \neq j, \sigma} t_{ij} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \quad (18.17)$$

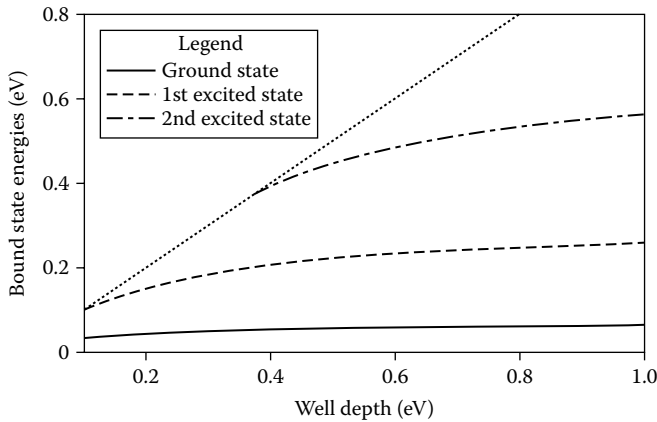


FIGURE 18.4 Variation of the bound state energies with the depth of the double quantum wells, $f_1 = f_2$. The straight line indicates where the bound state energy is equal to the well depth.

where the operators $\hat{c}_{i\sigma}^\dagger$ and $\hat{c}_{i\sigma}$ are the usual fermion creation and annihilation operators, and where i runs from 1 to $2n$ representing the two sites on each of the n dimers. The ϵ_i represent the potential energy at each site and the kinetic terms t_{ij} account for hopping from one site to another. There are two different kinds of interactions depending on whether an electron hops within a dimer, or between dimers. Hence, this kinetic parameter has two different values along the chain. Since there are no interactions between the states with $+\frac{1}{2}$ spin and those with $-\frac{1}{2}$ spin, the state space may be divided into two identical subspaces labeled by the electron spin. In each of these subspaces, the matrix corresponding to this chain of dimers is tri-diagonal and may be diagonalized to determine the itinerant eigenstates of the system. The coefficient t_d which represents the movement of an electron within a dimer causes the eigenstates to be split into two bands by an amount, $2t_d$. Within each of these bands, the coefficient t_l of the inter-dimer hopping, causes these two single electron bands to broaden each to about $2t_l$. Now, although the preceding description is qualitative, clearly whenever $t_d \sim t_l$, these bands will overlap and transitions between the states will be important.

Considering the band formation in the quantum well formulation, we can assign values to t_d and t_l . The two bands are separated by about 0.40 eV, so $t_d \sim 0.20$ eV, and using the bottom band to estimate t_l it widens until it is about 0.10 eV thick, so $t_l \sim 0.05$ eV.

The Hubbard model is the simplest second quantized model to capture all of the essential features of a real system of interacting particles [82]. In this model, electrons are the fundamental excitations. All states may be built up from the vacuum using the fermion creation operators. Once the number of sites is selected, n , and the number of electrons is determined, e , there are

$$\binom{2n}{e} = \frac{2n!}{e!(2n-e)!} \quad (18.18)$$

distinct electronic states. These states are built up by applying in normal order each of the combinations of e -electron states.

The second quantized Hamiltonian for the Hubbard model with electron–electron interactions is conventionally written as

$$\hat{H} = \sum_{i,\sigma} \epsilon_i \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} + \sum_{i \neq j, \sigma} t_{ij} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \sum_i U_i \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} \quad (18.19)$$

The last term accounts for electron–electron repulsion now that we have gone to a many electron system. In our simple model, this is limited to an onsite interaction. To this point, we have made each of the site energies equivalent ($\epsilon_i = \epsilon, \forall i$). In what follows, $U = 2.0$ eV which is the energy associated with two bare electrons lying about 0.7 nm apart. This choice seems reasonable within a well of width 2.0 nm. We try to solve the problem by considering the electron–electron term as a perturbation to study how it affects the electronic states of the dimer. The Hamiltonian in the absence of the final U -term may be diagonalized by the following unitary transformation:

$$\alpha_{1'\sigma}^\dagger = \frac{1}{\sqrt{2}}(c_{1\sigma}^\dagger + c_{2\sigma}^\dagger) \quad (18.20)$$

$$\alpha_{1'\sigma}^\dagger = \frac{1}{2\sqrt{2}}(c_{1\sigma}^\dagger - c_{2\sigma}^\dagger) \quad (18.21)$$

The linear combination of operators is simple because we have made the site energies ϵ_1 and ϵ_2 equal. These new operators are easily shown to obey the usual anticommutation relations for fermions and are said to be in the itinerant or distributed electron basis. The Hamiltonian may now be rewritten in terms of these new operators, as

$$\hat{H} = \sum_{\sigma} (\epsilon + t) \alpha_{1\sigma}^\dagger \alpha_{1\sigma} + \sum_{\sigma} (\epsilon - t) \alpha_{2\sigma}^\dagger \alpha_{2\sigma} + \frac{1}{2} \sum_{\sigma \neq \sigma'} V_{ijlm} \alpha_{i\sigma}^\dagger \alpha_{j\sigma'}^\dagger \alpha_{l\sigma'} \alpha_{m\sigma} \quad (18.22)$$

the vacuum state is represented by $|\Omega\rangle$. Acting with our Hamiltonian on this basis gives us the matrix form of the Hamiltonian:

$$\hat{H} = \begin{pmatrix} 2(\epsilon + t) + \frac{(U_1+U_2)}{4} & 0 & \frac{(U_1-U_2)}{4} & \frac{(U_2-U_1)}{4} & 0 & \frac{(U_1+U_2)}{4} \\ 0 & 2\epsilon & 0 & 0 & 0 & 0 \\ \frac{(U_1-U_2)}{4} & 0 & 2\epsilon + \frac{(U_1+U_2)}{4} & -\frac{(U_1+U_2)}{4} & 0 & \frac{(U_1-U_2)}{4} \\ \frac{(U_2-U_1)}{4} & 0 & -\frac{(U_1+U_2)}{4} & 2\epsilon + \frac{(U_1+U_2)}{4} & 0 & \frac{(U_2-U_1)}{4} \\ 0 & 0 & 0 & 0 & 2\epsilon & 0 \\ \frac{(U_1+U_2)}{4} & 0 & \frac{(U_1-U_2)}{4} & \frac{(U_2-U_1)}{4} & 0 & 2(\epsilon - t) + \frac{(U_1+U_2)}{4} \end{pmatrix}$$

The Hamiltonian is then diagonalized in order to determine its eigenstates. In the absence of the Coulomb repulsion ($U = 0$), the electrons are localized on the $2'$ site with energy $2(\epsilon - t)$, the gap to the first excited state being $2t$. However, as U is increased, the effect of the Coulomb interaction raises the energy of the ground state relative to the excited singlet states, which do not feel the Coulomb repulsion. Consequently, the gap between the ground state and the first excited state is actually reduced through the introduction, as shown in Figure 18.5. Thus, when conductivity is being considered, it is important to bear in mind that the gap is reduced by the presence of the electron–electron interactions.

To this point, we have made each of the site energies equivalent, ($\epsilon_i = \epsilon$), and the electron repulsion the same at all sites ($U_i = U$). The lone complication lies within the kinetic term, where the hopping parameter will depend on the two sites between which the electron hops. However, since we first consider the MT protofilament, it shall initially be a constant as well, $t = 0.4$ eV, where the choice comes from the separation of the energy bands in the 1D Schrödinger picture.

Since the Hamiltonian conserves both the particle number and the overall spin of each state, we are free to work in subspaces enumerated by the spin of the states it contains. This reduces the problem slightly since it is possible to consider individually the spin $+1$, spin 0 , and spin -1 systems separately in a two-electron system. In addition, the symmetry implies that the spin $+1$ and spin -1 systems have the same conductive properties. We also find that the ground state is always a state consisting of the lowest possible total spin.

When the inter-dimer hopping term is small relative to the intra-dimer hopping term $t_{inter} < t_{intra}$, the form of the ground state is independent of the number of dimers. In each case, the ground state is a linear combination of singlet states on each dimer. The singlet states are those where two electrons are

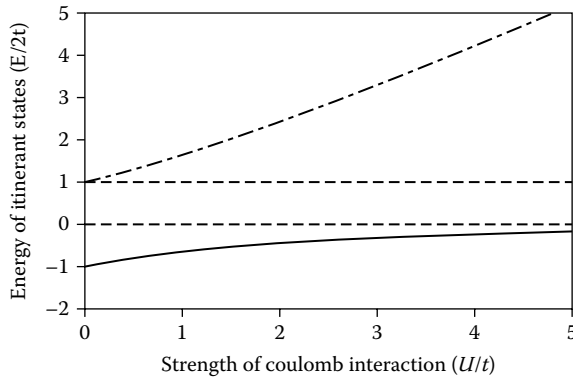


FIGURE 18.5 Energy of the itinerant states as a function of U/t with the energy plotted in units of $2t$.

paired on an individual binding site, or where the state consists of the anti-symmetric combination of an up-electron and a down-electron on neighboring binding sites of the dimer. When the inter-dimer hopping parameter is increased, the density of states becomes more uniform between the ground state and the highest energy eigenstate. With the gap between the ground state and the first excited state of the system becoming diminished. Since the hopping term within a dimer is largest, the system prefers to have two electrons with the same spin within the same dimer in the ground state.

For electronic conductivity, the band formation importantly depends upon the hopping parameter between dimers. When this parameter is small relative to the intra-dimer hopping, $t_{inter}/t_{intra} \sim 0.1$, there remains a rather large gap between the ground state and the first excited state of about $2t_{intra}$ which is reduced as the repulsion term increases. However, when the hopping between dimers is equal to the hopping within dimers, $t_{inter} \sim t_{intra}$, the gap is minimized. In the event that $t_{inter} > t_{intra}$, we would effectively have to relabel the dimers and would then return to the picture where $t_{inter} < t_{intra}$.

Kubo was the first to derive formulae for the electrical conductivity in solids, and his derivation is applied to our case in the present section [77]. In the Coulomb gauge, we write the vector potential for a uniform electric field along the direction of the chain, $A(t)$, as

$$A(t) = \frac{cE(t)}{-i\omega} e^{i\omega t}, \quad (18.23)$$

Adding a field-coupling term to the Hubbard Hamiltonian and using a canonical transformation gives [91]

$$\hat{H}'_0 = -t \sum_{i\sigma} [\hat{c}_{i\sigma}^\dagger \hat{c}_{i+1\sigma} e^{-ieA(t)/c} + \hat{c}_{i\sigma}^\dagger \hat{c}_{i-1\sigma} e^{ieA(t)/c}]. \quad (18.24)$$

where a is the lattice spacing and c is the speed of light. Expanding to second order in the vector potential, we find

$$\hat{H}_A = \frac{A}{c} \hat{j} - e \frac{A^2}{c^2} a^2 \hat{H}_0, \quad (18.25)$$

where

$$\hat{j} = ieat \sum_{i\sigma} (\hat{c}_{i\sigma}^\dagger \hat{c}_{i+1\sigma} - \hat{c}_{i\sigma}^\dagger \hat{c}_{i-1\sigma}). \quad (18.26)$$

So, the energy of the original system is corrected to this order and the interaction term is proportional to the current.

Suppose a time-dependent external electric field is applied to a solid:

$$E_\alpha^{\text{ext}}(r, t) = \Xi_\alpha^{\text{ext}} e^{iq \cdot r - i\omega t}, \quad (18.27)$$

where α represents the Cartesian coordinate directions. In a linear response theory, the induced current is proportional to the applied electric field:

$$J_\alpha(r, t) = \sum_\beta \sigma'_{\alpha\beta}(q, \omega) \Xi_\beta^{\text{ext}} e^{iq \cdot r - \omega t}, \quad (18.28)$$

where $\sigma'_{\alpha\beta}$ is a parameter relating the observed current density to the applied field's direction, its periodicity, and frequency. However, the symbol appearing in (18.28) is not the conductivity we seek. Rather, we want the conductivity which represents the response to the total electric field in the solid, a quantity that can be measured. This conductivity takes into account all of the currents, induced by the external fields, that

create their own electric fields. Thus, we seek $\sigma_{\alpha\beta}$ that relates the macroscopic electric field to the currents of the system.

$$J_{\alpha}(r, t) = \sum_{\beta} \sigma_{\alpha\beta}(q, \omega) E_{\beta} e^{iq \cdot r - i\omega t} \quad (18.29)$$

$$E_{\alpha}(r, t) = \Xi_{\alpha} e^{iq \cdot r - i\omega t} \quad (18.30)$$

$$\sigma_{\alpha\beta} = \text{Re}(\sigma_{\alpha\beta}) + i \text{Im}(\sigma_{\alpha\beta}) \quad (18.31)$$

We write the Hamiltonian of the system as $\hat{H} + \hat{H}'$ where the latter term contains the electric field, which we shall introduce as a time-dependent perturbation. The evolution of the operators is given as follows [30]:

$$\hat{j}(x, t) = e^{i\hat{H}t} \hat{j}(x) e^{-i\hat{H}t}. \quad (18.32)$$

where $\hat{H}$ is the unperturbed Hamiltonian. Thus, the interaction picture is adopted. The Kubo formula for electrical conductivity gives the result in terms of a current–current correlation function:

$$\sigma_{\alpha\beta}(q, \omega) = \frac{1}{\omega} \int_0^{\infty} dt e^{i\omega t} \langle \Omega | \hat{j}_{\alpha}^{\dagger}(q, t) \hat{j}_{\beta}(q, 0) - \hat{j}_{\beta}(q, 0) \hat{j}_{\alpha}^{\dagger}(q, t) | \Omega \rangle + i \frac{n_0 e^2}{m\omega} \delta_{\alpha\beta} \quad (18.33)$$

where $|\Omega\rangle$ is the ground state of the system, which has not been perturbed by the electric field. The first term in (18.33) is known as the incoherent contribution to the conductivity. Now, we use (18.32) in the preceding equation and use the fact that the basis we are using is that of the eigenstates. Consequently, with proper normalization the new expression for the real part of the conductivity is [90]

$$\sigma(\omega) = \frac{\pi}{Z} \frac{(1 - e^{-\beta\omega})}{\omega} \sum_{n,m} e^{-\beta E_m} | \langle n | \hat{j}_{\alpha} | m \rangle |^2 \delta[\omega - (E_n - E_m)] \quad (18.34)$$

The DC conductivity is calculated by a procedure [90,91], which employs the sum rule

$$\int_0^{\infty} \sigma'(\omega) d\omega = -\frac{\pi e^2}{2} a^2 \langle K \rangle \quad (18.35)$$

where

$$\sigma'(\omega) = D\delta(\omega) + \sigma_{kubo}(\omega) \quad (18.36)$$

and $\langle K \rangle$ is the expectation value of the kinetic energy term of the Hamiltonian. The strategy of computing $\sigma(\omega)$, and subsequently the Drude contribution, at either zero or finite temperature is reasonably straight forward. The first step is to construct the matrix corresponding to the Hamiltonian [Equation (18.19)] in a subspace which is defined by the geometry of the lattice, the number of electrons, and the total electron spin. This leads to serious computational difficulties since for a moderately sized lattice of say ten dimers and eight electrons of arbitrary spin, the corresponding space is about 77 million states. A matrix of dimension 77 million square has about 5.9×10^{15} elements. Obviously, working on a moderately sized lattice, we are restricted to a low concentration of electrons. Due to electron-hole symmetry, we are also able to study the nearly filled situation. It is useful to keep in mind the relationship

$$\lambda = \frac{1240 \text{ nm} \cdot \text{eV}}{\text{Energy}}. \quad (18.37)$$

So that in the plots following, the visible light range is between 1.75 eV and 3.10 eV. Energies larger than this range correspond to ultraviolet, and those smaller to infrared wavelengths. Since cells respond to light in the near-infrared [4], and centrioles comprised of MTs are proposed to be the site of light detection [5], absorption in this energy range is of great interest to cell biology. These energies should also be compared

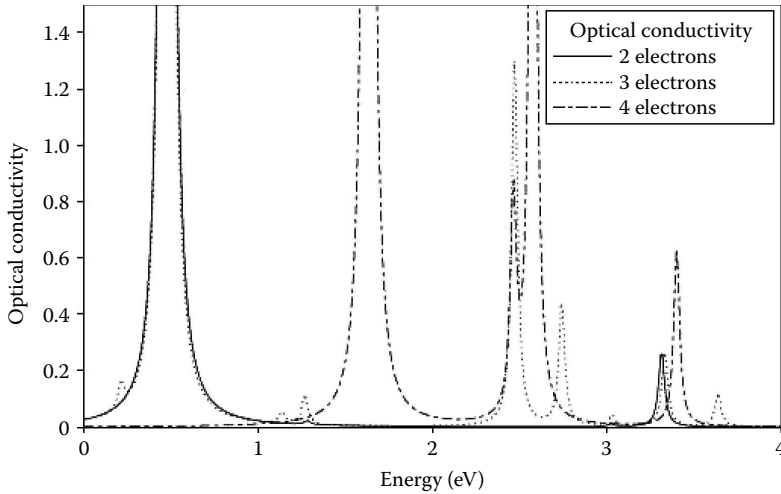


FIGURE 18.6 Optical conductivity of the two dimer system with two, three, and four electrons.

to the energy available from ATP and GTP hydrolysis, which are 0.49 eV and 0.22 eV, respectively. These values provide limits for the interactions of chemical energy in the conduction process.

At zero temperature, the linear polymer results are summarized concisely by saying that there is no DC conductivity that agrees with the previous modeling of Hubbard systems [44].

The following results for the 1D chain of tubulin dimers represent the AC conductivity. There are peaks in $\sigma(\omega)$, also known as the optical conductivity, which correspond to excitations between electronic states of the polymer. Electronic excitation results in electron redistribution and hence the development of a current. We have selected $\epsilon = 0.02$ eV to present our results, which is roughly kT and is smaller than the three energy scales in the Hubbard model, t , U , and $4t^2/U$ [82] so we expect to be able to pick out features corresponding to specific transitions in the predicted optical spectra.

We shall begin with two dimers and $t = 0.4$ eV, $U = 2.0$ eV, and equal site energy on the α and β monomer sites. This system can support up to eight electrons. The results for these systems are presented in Figure 18.6. The predicted spectra for the two and three electron cases are quite similar. The differences include a new but small peak in the three electron spectrum at an energy just below the major peak, and variations in the detail of the absorption band centered about 2.6 eV. However, the qualitative similarity of the two spectra is remarkable in terms of band locations. Contrast these two cases with the four-electron system that corresponds to half-filling. The difference is that the peak in the optical conductivity spectrum increases dramatically from about 0.48 eV to 1.64 eV. This change indicates how the half-filled case is special. The ground state in this system largely consists of electrons spread out to minimize the Coulombic energy. Exciting this system by moving any particular electron will come at the cost of greatly increasing the electron–electron repulsion and, consequently, the location of the required excitation energy corresponds roughly to U .

As the length of the polymer is increased, results remain qualitatively similar. In the three-dimer structure, results are shown for the conductivity in Figure 18.7(a) and for the integrated conductivity in Figure 18.7(b). Once again, the spectra are quite similar for all fillings aside from the half-filled case which consists of six electrons. The shift of the conductivity to higher energies can be clearly seen as the filling fraction increases towards half filling. We now present results where the broadening parameter, ϵ , and the temperature have been varied to see the effect that they have on the results. Figure 18.8 demonstrates that as the width is decreased, the spectrum becomes sharper. Note how the two peaks in this simple spectrum that occur near 2.5 eV are successively blended into a band as the ϵ is increased. Peaks represent transitions from one specific electron configuration to another and are weighted with the associated conductivity.

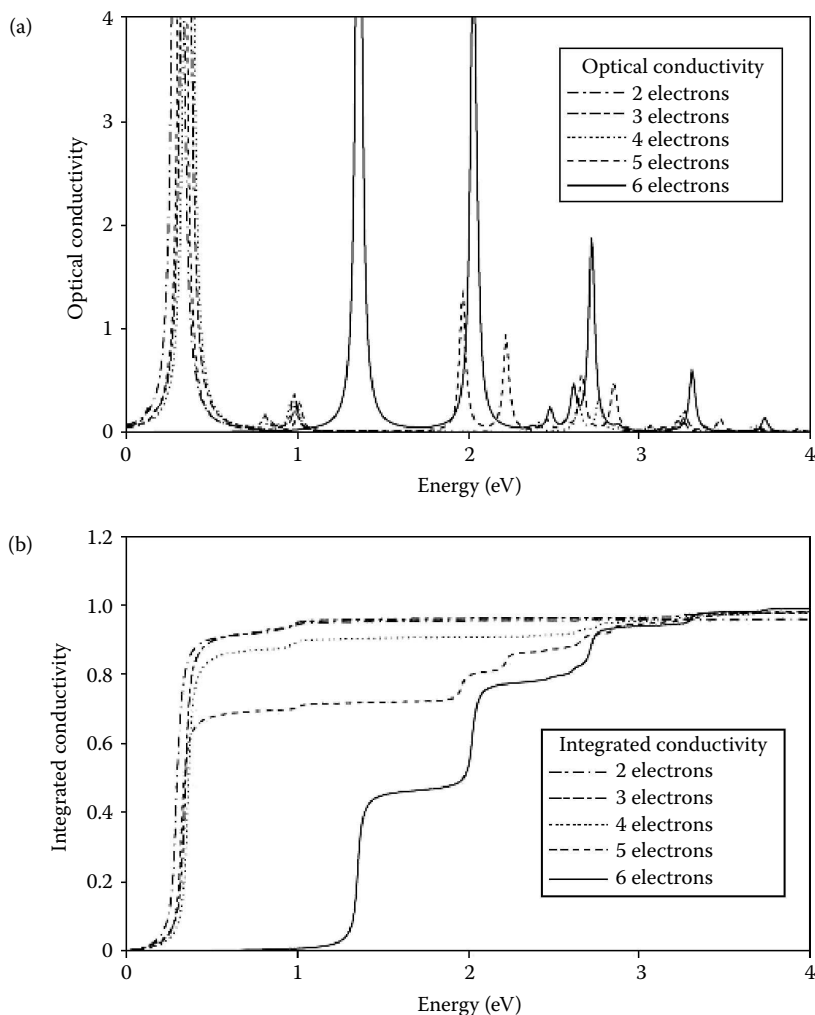


FIGURE 18.7 (a) Optical conductivity of the three-dimer system from two electrons through half filling. (b) Integrated optical conductivity of the three-dimer system shows that indeed the protofilament is an insulator. The gap for excitation of the half-filled polymer is seen easily here.

The effect of changing the temperature is to excite electron configurations other than the ground state. Consequently, transitions from one excited state to another can now contribute to the conductivity. Figure 18.9 shows how this often reduces the gap to conduction: all the major peaks in the 300 K spectrum shift slightly towards lower energies in the 3000 K spectrum, but many more features develop as well as states that previously did not contribute to conductivity may make transitions to excited states. Physiological temperature of 300 K ($kT = 0.026$ eV) is small relative to the system's energy scales, consequently lowering the temperature further causes very little change to the spectrum.

If the α -tubulin and β -tubulin sites have different energies, then the MT A and B lattices can be distinguished. The effect of this change on the optical spectrum seems to be small when individual protofilaments are considered. In addition, the sum rule shows that protofilaments remain insulators. There are a few parameters we can consider individually and which can be compared as the size of the system changes: the threshold to conductivity, and the expectation value of the kinetic energy operator. The threshold to conductivity is simply the location of the first absorption peak in the optical spectrum.

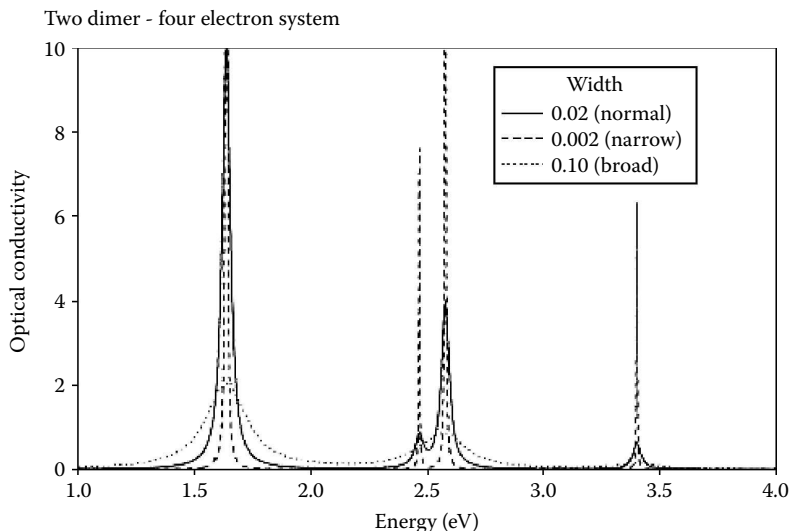


FIGURE 18.8 Optical conductivity spectrum variation with the broadening parameter.

It is a decreasing function of the polymer length. Starting at 0.48 eV for the two-dimer system, it seems to drop a little faster than $1/n$ for systems consisting of n dimers and containing two electrons. The fitted value in Figure 18.10 varies $\propto 1/n^{1.4}$. Extrapolating to larger values of n , this gap appears certain to fall into the range where these excited states could be thermally excited. The protofilament then develops a finite DC conductivity at finite temperature. It is interesting that although the first peak does move towards zero, the fraction of the conductivity it contains is gradually diminished. Some of the conduction continues to reside in a peak located close to t in the energy spectrum. In the 2D lattice, there is a phase transition in the dimensionality. DC conduction becomes possible even at zero temperature. However, we shall still be concerned about the boundary conditions since calculations are being performed on a small lattice.

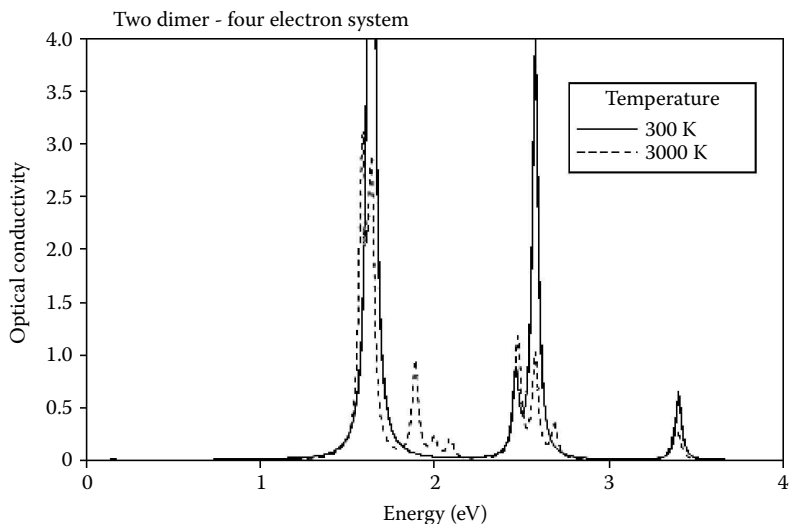


FIGURE 18.9 The variation of the optical conductivity spectrum with temperature is shown.

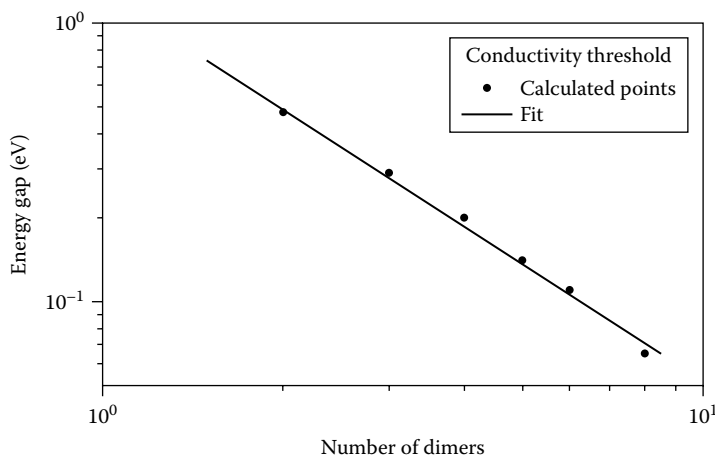


FIGURE 18.10 Threshold to conduction as a function of protofilament length.

Consider the kinetic energy/electron that has been calculated in a Hubbard model with our parameters for t and U and with either one, two or three protofilaments. Figure 18.11 demonstrates how the kinetic energy of each electron approaches the limit of $2t$ or 0.80 eV as the size of the system increases. We can easily see that for more than about 20 electron sites, the average kinetic energy of each electron starts to approach the infinite limit and we expect that the results obtained from our model should converge to the long protofilament limit in a similar manner. In particular, we are considering 28 electron sites in our MT model, which span three protofilaments. While this is not the ideal situation, the boundary conditions are not expected to create huge effects in our results. We are also using open boundary conditions since they minimize the boundary effects when compared with a variety of periodic boundary conditions.

The unit cells depicted in Figure 18.12 are the basis for the calculations. In the A lattice used for the calculations, the α -dimers connect to β -dimers along the $\bar{3}$ -start helix, both within the lattice and at the seam when the MT is wrapped up. In the B lattice, the connections are $\alpha - \alpha$ or $\beta - \beta$ along the $\bar{3}$ -start helix and never $\alpha - \beta$. For comparison, calculations have also been carried out with a flat tubulin sheet, specifically an A lattice MT lacking periodic boundary conditions in the lateral direction. This simulates the

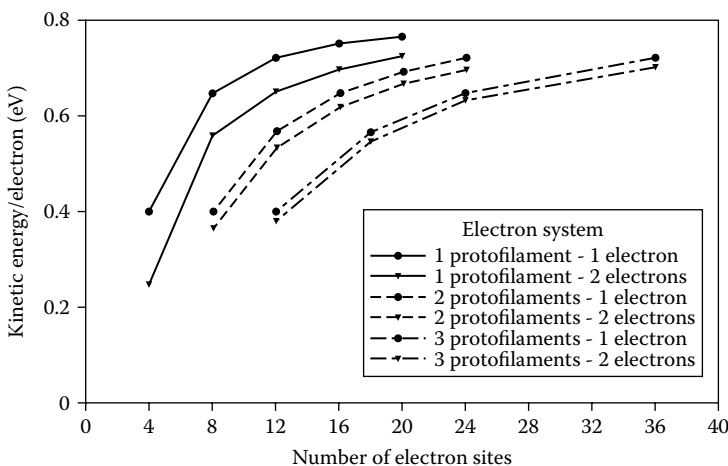


FIGURE 18.11 Kinetic energy per electron as a function of system size.

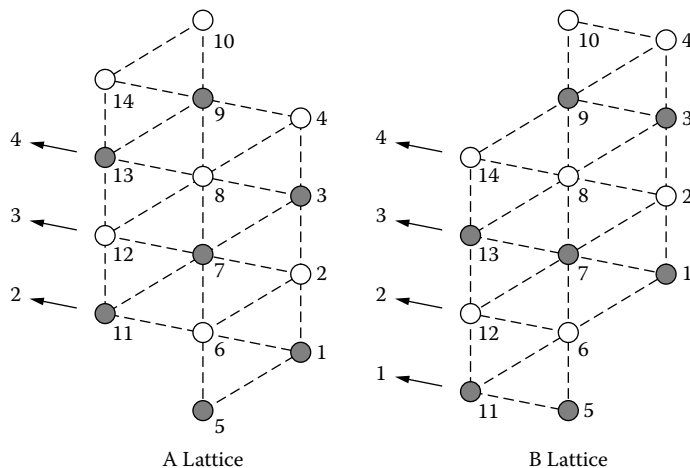


FIGURE 18.12 Lattice unit cells for conductivity calculations. The unit cell on the left corresponds to the MT A lattice, while that on the right corresponds to the MT B lattice.

MT when it has been unwrapped. We recall that the MT lattice is triangular and has three distinct hopping directions. Since protofilaments are strongly bound together along their length but relatively weakly bound to other protofilaments, one also expects the hopping parameter to be much smaller when an electron moves between protofilaments. The calculations have been carried out with the same parameters as in the one-dimensional case, $U = 2.00$ eV, $t = 0.40$ eV along the protofilaments and the site energies have been set equal to zero for all sites. The value of the hopping parameter along the $\bar{3}$ -start helix is called t_ℓ (left) and that along the 8-start helix that connects diagonally from a monomer in a direction up and to the right is called t_r (right).

We consider first the results of the MT lattice which has not rolled up to form a tube but is that of a tubulin sheet where $t_r = 0$ and only the value of t_ℓ is varied (see Figure 18.13). Since $t_r = 0$, the nature of the lattice remains rectangular to this point. When $t_\ell = 0$, this particular system is equivalent to individual protofilaments. The two peaks in the optical spectrum arise from the fact that the protofilaments have different lengths. The absorption peaks of 0.22 eV and 0.40 eV correspond roughly to those of the two-dimer and three-dimer, two electron cases discussed earlier in the chapter. The difference here is that the two electrons are spread over three protofilaments and consequently the effect of the electron repulsion is reduced. However, as t_ℓ is increased from zero, the entire lattice becomes accessible and the protofilament character of the optical spectrum is lost. The two peaks of the t_ℓ spectrum coalesce into a single peak with a large activation energy. A second but smaller absorption peak forms just above 0.50 eV. In the plot of the integrated conductivity, we can see that there is also a smaller peak at 0.16 eV when $t_\ell = 0.10$ eV. The source of this peak is an absorption for conduction along the direction of t_ℓ . When t_ℓ is increased to 0.40 eV, this peak occurs at 0.61 eV. Thus, this peak seems to occur at roughly $1.5t_\ell$. The activation energy for conduction along the protofilament axis also increases but is not as sensitive to t_ℓ . Finally, examination of the integrated conductivity demonstrates that the total absorption combines such that the Drude weight remains zero.

The results in Figure 18.14 show a difference between an individual protofilament and a tubulin sheet. The behavior here is inconsistent with that of an insulator since the Drude weight is non-zero. In the optical conductivity, we find that the largest peak occurs at the same location in all directions but with different weightings. The first line traces the results for $t_r = 0.00$ eV and has peaks at 0.38 eV and 0.61 eV. Once the third hopping direction is allowed, the peak absorption is raised slightly to 0.43 eV for $t_r = 0.10$ eV and a smaller peak forms at 0.74 eV. The lower peak consists mainly of hopping along the protofilament while the second peak consists largely of hopping along the t_ℓ direction, which also corresponds to a large

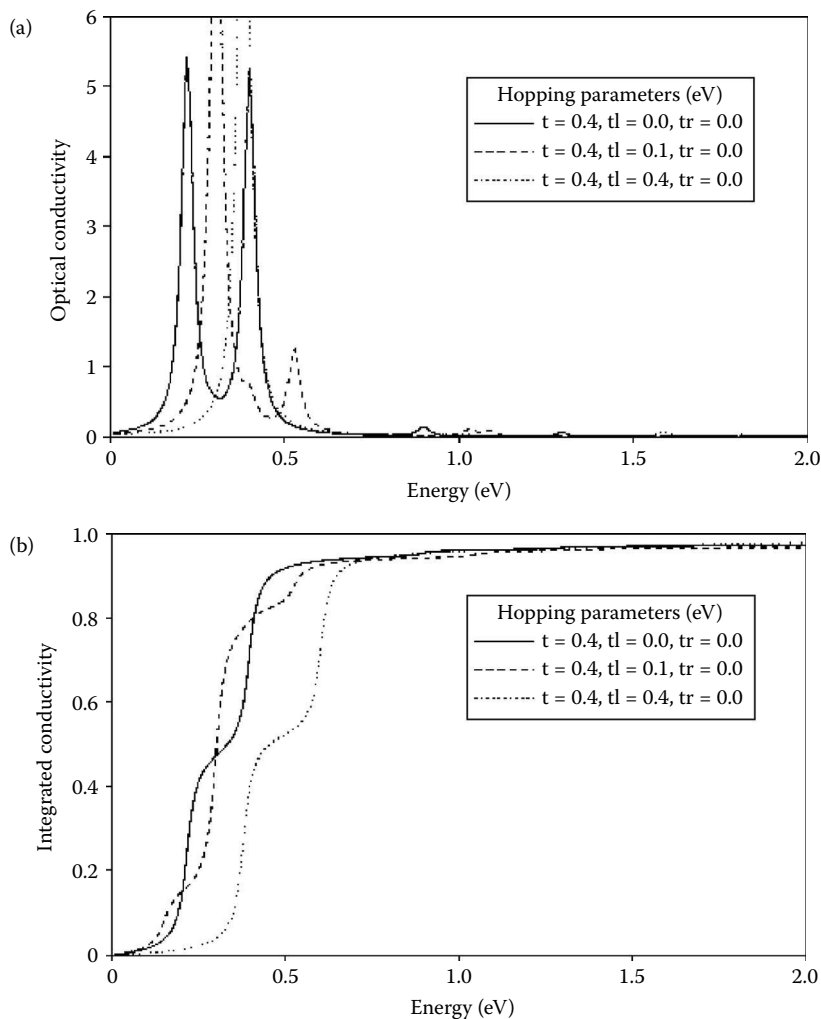


FIGURE 18.13 (a) Optical conductivity along the protofilament axis of the tubulin sheet consisting of a 14-site lattice with two electrons while $t_r = 0$ and t_l is varied. (b) Integrated conductivity of the tubulin sheet with two electrons while $t_r = 0$ and t_l is varied.

hopping parameter. Once t_r is raised to 0.40 eV so that it has the same magnitude as the other hopping parameters, there is only a single peak at 0.70 eV.

The effect of applying the MT A lattice boundary conditions to our system is shown in Figure 18.15, where we see that changing only t_l to a finite quantity while still maintaining t_r as zero is sufficient to result in a non-zero Drude weight. Thus, the periodic boundary conditions in the lateral direction seem to act in a manner similar to the additional hopping directions of the unwrapped lattice. When the kinetic parameters between protofilaments are set to zero, the optical conductivity spectrum shows there are two large absorption peaks at about 0.23 eV and 0.40 eV. As t_l increased, these peaks move higher in energy and the second peak becomes smaller in integrated weight. Eventually, once $t_l = t$, there is a single large peak near 0.47 eV. Introduction of the third hopping direction serves only to raise further the absorption peak, up to 0.70 eV in the case where the hopping parameter is 0.40 eV in all directions.

From Figure 18.15 one can see that more conduction results in the B lattice for a small t_l , such as 0.1 eV, than in the MT A lattice, by comparing it with Figure 18.16. However, even the positioning of absorption

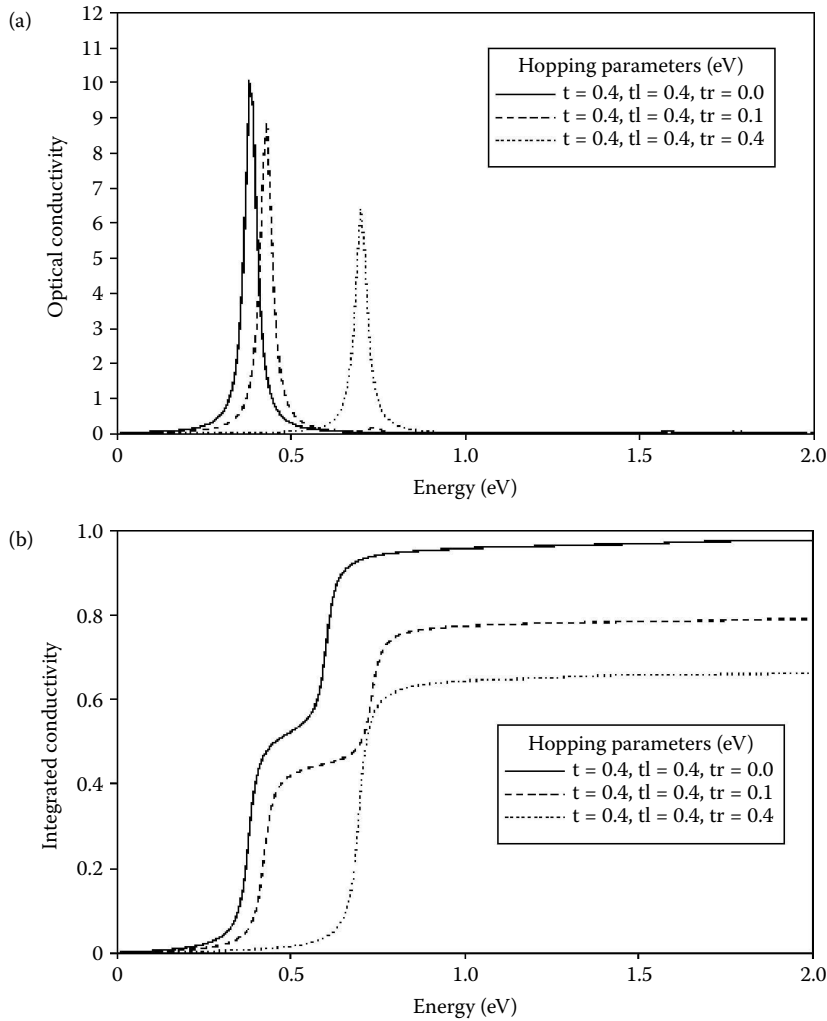


FIGURE 18.14 (a) Optical conductivity along the protofilament axis of the unwrapped 14-site lattice with two electrons while $t_l = 0.4$ and t_r is varied. (b) Integrated conductivity of the unwrapped 14-site lattice with two electrons while $t_l = 0.4$ and t_r is varied.

peaks is similar in the two lattices. Again, it is sufficient for only one of t_l or t_r to be non-zero for the Drude weight to be finite.

Finally, we compare the MT A and B lattices with the unwrapped tubulin sheet as well as with individual protofilaments of tubulin dimers. There is an interesting geometrical interpretation to the results, which shows that the MT A and MT B lattices have identical conduction properties when $t_l = t_r$. This is because in our calculations we have considered all monomers to have the same site energy. Consequently, the B lattice can be viewed as an A lattice with the opposite helicity but the same structure. The sign of the system's helicity does not affect the conduction properties along the major axis. The unwrapped MT lattice is not as good a conductor since some of the hopping freedom has been removed from the lattice but does remain conducting, provided that both t_l and t_r are non-zero.

If we consider the case where one of t_l or t_r is quite small, however, the wrapping of the lattice is much more important. As discussed earlier, the unwrapped tubulin sheet is an insulator while the MT A and B lattices may carry electrons. This observation is especially interesting given that MTs have been observed to zip up and essentially change their structure from a tube to that of a sheet *in vivo*.

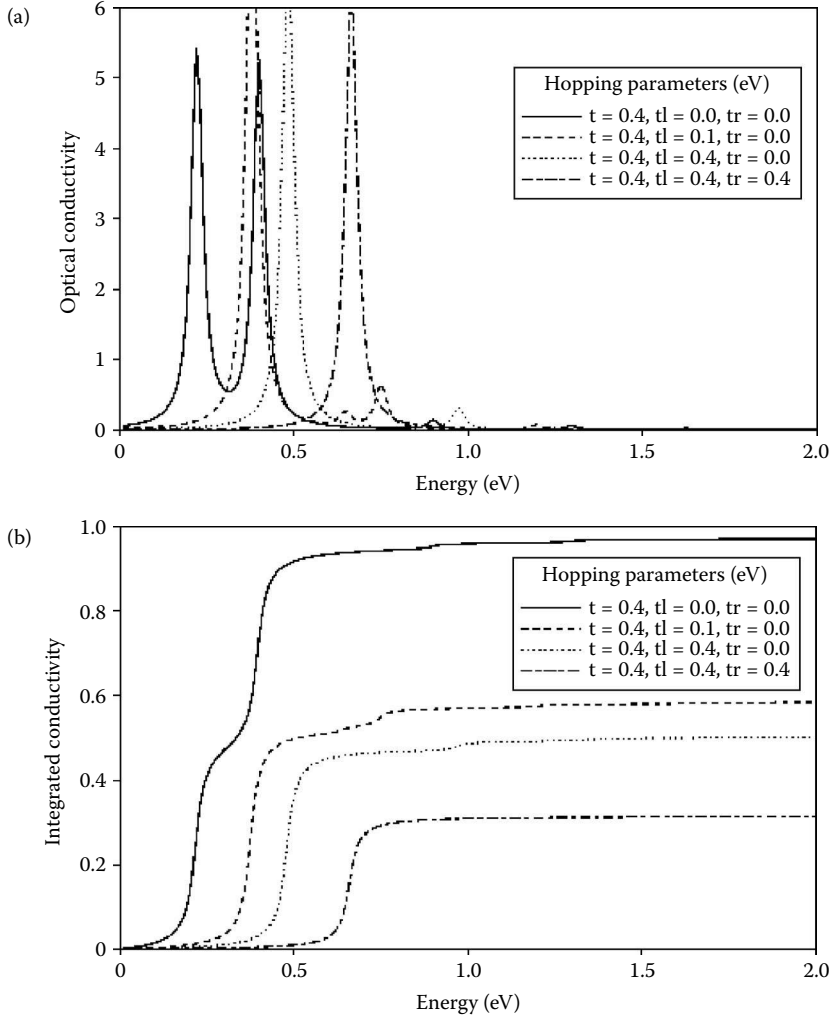


FIGURE 18.15 (a) Optical conductivity along the protofilament direction of the 14-site MT B lattice with two electrons. (b) Integrated conductivity of the 14-site MT B with two electrons shows a non-zero weight for finite t_l .

Our calculations based on the Hubbard model of electron hopping show that the MT is not an insulator, but that instead its conductivity depends on the lattice geometry and whether the MT is wrapped up or not. How well or poorly does one expect the MT to conduct relative to known semi-conductors? Referring back to Equations (18.35) and (18.36), the DC conductivity of the MT is simply given by D as

$$\frac{D}{2} = -\frac{\pi e^2}{2} a^2 < K > - \int_0^\infty \sigma_{kubo}(\omega) d\omega. \quad (18.38)$$

In this equation, the constants can be factored, and what remains is the fraction that can be read from the graphs presented earlier. The Kubo fraction (KF) is the limit of the integral as the frequency tends to infinity.

$$\frac{D}{2} = -\frac{\pi e^2}{2} a^2 < K > (1 - KF). \quad (18.39)$$

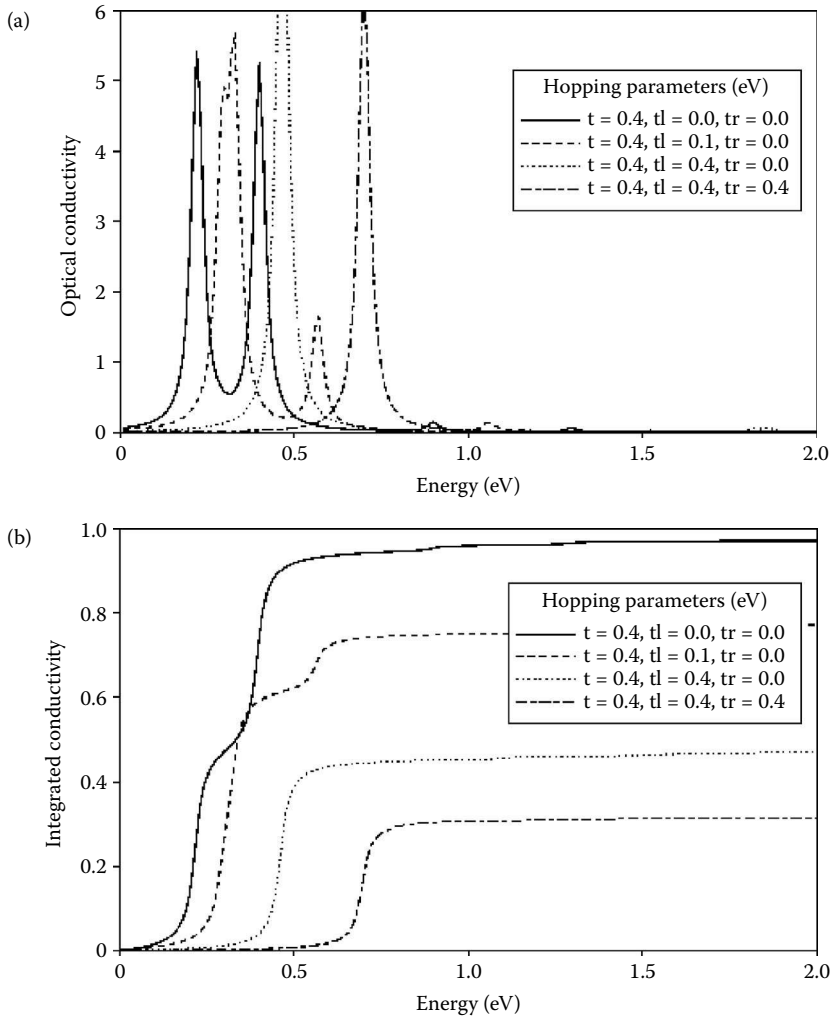


FIGURE 18.16 (a) Optical conductivity along the protofilament direction of the 14-site MT A lattice with two electrons. (b) Integrated conductivity of the 14-site MT A lattice with two electrons shows a non-zero Drude weight for finite t_l .

By returning constants of the system so that the dimensionality of D is correct, we find

$$D = -\frac{\pi n e^2 a^2}{\hbar} \frac{\langle K \rangle}{t} (1 - KF). \quad (18.40)$$

Finally, we are in a position to predict the conductivity of individual MTs of varying lattice types. For the following calculations, the electron density has been taken to be of two electrons within the lattice of 14 dimers and corresponds to a density of $1.8 \times 10^{18}/\text{cm}^3$. Thus, the results must assume that the conductivity remains constant as the length of the polymer is extended, provided the degree of filling remains the same.

The values in Table 18.1 should be compared with the conductivity of metals, such as copper ($6 \times 10^7 \Omega\text{m}^{-1}$) and iron ($1 \times 10^7 \Omega\text{m}^{-1}$). Indeed the values can also be compared to the intrinsic semiconductors germanium ($2.5 \Omega\text{m}^{-1}$) and silicon ($4 \times 10^{-4} \Omega\text{m}^{-1}$). Given these comparisons, the MT may indeed be quite a good semi-conductor given our assumptions. What is particularly interesting is the

TABLE 18.1 Calculated Conductivity and Resistance of a 1- μm Polymer Composed of Tubulin Dimers

Tubulin Structure	Conductivity ($\Omega^{-1}m^{-1}$)	Resistance
Individual Protofilament	Non-Conducting	
Tubulin sheet ($t_l = t_r = 0.04 \text{ eV}$)	9.4×10^3	418k Ω
Tubulin sheet ($t_l = t_r = 0.1 \text{ eV}$)	2.0×10^4	195k Ω
Tubulin sheet ($t_l = t_r = 0.4 \text{ eV}$)	6.3×10^4	62k Ω
MT (A lattice, $t_l = t_r = 0.04 \text{ eV}$)	1.8×10^4	219k Ω
MT (A lattice, $t_l = t_r = 0.1 \text{ eV}$)	4.5×10^4	87k Ω
MT (A lattice, $t_l = t_r = 0.4 \text{ eV}$)	1.6×10^5	25k Ω
MT (B lattice, $t_l = t_r = 0.04 \text{ eV}$)	1.9×10^4	211k Ω
MT (B lattice, $t_l = t_r = 0.1 \text{ eV}$)	4.7×10^4	84k Ω
MT (B lattice, $t_l = t_r = 0.4 \text{ eV}$)	1.6×10^5	24k Ω

way the conduction properties depend on the lattice and the particular boundary conditions. The situation can be compared to carbon nanotubes which have a similar size and structure to MTs. In addition, theoretical consideration of nanotube structure predicts that the conduction properties depend on the boundary conditions. In the case of nanotubes, this means the way in which the graphene sheet is wrapped up to form the nanotube. It is quite interesting that particular sets of boundary conditions produce a semi-conducting nanotube while the appropriate choice of wrapping the nanotube gives rise to metallic conduction [31].

The ability to change from an insulator to a conductor by a simple geometrical change could be biologically relevant even without observing a single electron to be conducted along the MT since it is the conductivity property that affects the way the cell views and responds to external electromagnetic fields. Specifically, the reflectivity of MTs to electromagnetic fields is high when they are conducting, and, consequently, they could act to direct infrared signals to the interior of the centrosome. Since the centrioles are maintained at right angles to each other, the cell would be able to determine the location in latitude and longitude of a light source if the MTs are conducting [4,5]. Peaks in the AC conductivity are all at energies significantly above the thermal activation threshold so no conductivity is expected along a darkened MT, but photoconduction in the infrared-visible range is possible. Therefore, it is possible this photo-activated conduction has functional repercussions. In addition to the intrinsic semiconducting behavior of MTs, and possibly AFs, these protein filaments possess very intensely active cable properties when immersed in an ionic buffer solution. In the next section, we discuss the ionic conductivity effects of AFs.

18.6 Actin Filaments Support Nonlinear Ionic Waves

Several experiments [12,83] indicate the possibility of ionic wave generation along AFs. As the condensed cloud of counter-ions separates the filament core from the rest of the ions in the bulk solution, we expect it to act as a dielectric medium between the two. It has both resistive and capacitive components associated with each monomer that makes up the AF. Ion flow is expected to occur at a radial distance from the surface of the filament approximately equal to the Bjerrum length. An inductive component is proposed to emerge due to the actin's double-stranded helical structure that induces the ionic flow in a solenoidal manner. Due to the presence of the sheath of counter-ions around the AF, these polymers act as biological "electrical wires" [83], and have been modeled as nonlinear inhomogeneous transmission lines propagating nonlinear dispersive solitary waves. Recently, Lader *et al.* [78] applied an input voltage pulse with an amplitude of approximately 200 mV and a duration of 800 μs to an AF, and measured electrical signals at the opposite end of the AF, indicating that AFs support ionic waves in the form of axial nonlinear currents. In an earlier experiment [83], the wave patterns observed in electrically stimulated single AFs were remarkably similar to recorded solitary waveforms for electrically stimulated nonlinear transmission lines [85]. Considering the AFs highly nonlinear complex physical structure and the thermal fluctuations of the counter-ionic

cloud [105], the observation of soliton-like ionic waves is consistent with the idea of AFs functioning as biological transmission lines.

The electro-conductive medium is a condensed cloud of ions surrounding the polymer and separated from it due to the thermal fluctuations in the solution. The distance beyond which thermal fluctuations are stronger than the electrostatic attractions or repulsions between charges in solution is defined as the so-called Bjerrum length, λ_B . With the dielectric constant of the medium denoted by ε , the Bjerrum length is given by

$$\frac{e^2}{4\pi\varepsilon\varepsilon_0\lambda_B} = k_B T \quad (18.41)$$

for a given temperature T in Kelvin. Here e is the electronic charge, ε the dielectric constant, ε_0 the permittivity of the vacuum, and k_B is Boltzmann's constant. For a temperature of 293 K, we find that $k_B = 7.13 \times 10^{-10} m$. Counter-ion condensation occurs when the mean distance between charges, b , is such that $\lambda_B b/b = S > 1$. Each actin monomer carries an excess of 14 negative charges in vacuum, and accounting for events such as protonation of histidines, and assuming there to be 3 histidines per actin monomer, there exist 11 fundamental charges per actin subunit [143]. Assuming an average of 370 monomers per μm , we find there is approximately $4e/nm$. Thus, we expect a linear charge spacing of $b = 2.5 \times 10^{-10} m$ so $S = 2.85$. As the effective charge, q_{eff} , or renormalized rod charge is the bare value divided by S , we find $q_{eff} = 3.93e/monomer$. Consequently, it can be shown that approximately 99% of the counter-ion population is predominantly constrained within a radius of $8nm$ [110] around the polymer's radial axis [158]. Significant ionic movements within this "tightly bound" ionic cloud are therefore allowed along the length of the actin, provided that it is shielded from the bulk solution [104,107].

The physical significance of each of the components of the electrical network, and additional details, are described in [148]. For actin in solution, a key feature is that the positively charged end assembles more quickly than the negatively charged end [135]. This results in an asymmetry in the charges at the ends of the filaments and F-actin's electric polarization. Actin monomers arrange themselves head to head to form actin dimers resulting in an alternating distribution of electric dipole moments along the length of the filament [73]. We assume, therefore, that there is a helical distribution of ions winding around the filament at approximately one Bjerrum length. This corresponds to a solenoid in which a fluctuating current flows as a result of voltage gradient between the two ends.

The capacitive element of the electric circuit is obtained following the observation of oppositely charged layers surrounding the filament surface. We envisage the protein surface's negative charge to be distributed homogeneously on a cylinder defining the filament surface. Furthermore, positive counter-ionic charges in the bulk are expected to form another cylinder at a radius greater than the AF itself, approximately one Bjerrum length, λ_B , away from the actin surface, which includes the condensed ions. The permittivity, ε_0 , is given by $\varepsilon = \varepsilon_0\varepsilon_r$ where ε_r is the relative permittivity which we take to be that of water, i.e., $\varepsilon_r = 80$. We take the length of an actin monomer typically as $a = 5.4 nm$ and the radius of the actin filament, r_{actin} , to be $r_{actin} = 2.5 nm$ [14]. The next step is to consider a cylindrical Gaussian surface of length a whose radius is r such that $r_{actin} < r < r_{actin} + \lambda_B$.

Applying of Gauss's law for the total charge enclosed in the cylinder, we have the following expression for the capacitance

$$C_0 = \frac{2\pi\varepsilon a}{\ln\left(\frac{r_{actin} + \lambda_B}{r_{actin}}\right)} \quad (18.42)$$

With the parameters given previously, we estimate the capacitance per monomer is $C_0 = 96 \times 10^{-6} pF$. The resistive part is obtained from Ohm's law. Taking into account the potential difference and the current I , the magnitude of the resistance, $R = V/I$, for an actin filament is given by

$$R = \frac{\rho \ln\left(\frac{r_{actin} + \lambda_B}{r_{actin}}\right)}{2\pi l} \quad (18.43)$$

where ρ is the resistivity. Typically, for K^+ and Na^+ , intracellular ionic concentrations are 0.15 M and 0.02 M , respectively [147]. Kohlrausch's law states that the molar conductance of a salt solution is the sum of the conductivities of the ions comprising the salt solution. Thus

$$\sigma = \Lambda_0^{K^+} c_{K^+} + \Lambda_0^{Na^+} c_{Na^+} = 1.21(\Omega m)^{-1} \quad (18.44)$$

Using this with $\rho = \sigma^{-1}$, the resistance estimate becomes $R = 6.11 \text{ } M\Omega$, which is much lower than pure water since $R_{\text{water}} = 1.8 \times 10^6 \text{ } M\Omega$.

To describe the properties of the whole filament, we simply connect n subcircuits, as described earlier, to obtain an effective resistance, inductance, and capacitance, respectively, such that:

$$R_{\text{eff}} = \left(\sum_{i=1}^n \frac{1}{R_{2,i}} \right)^{-1} + \sum_{i=1}^n R_{1,i} \quad (18.45)$$

$$L_{\text{eff}} = \sum_{i=1}^n L_i \quad (18.46)$$

and

$$C_{\text{eff}} = \sum_{i=1}^n C_{0,i} \quad (18.47)$$

where $R_{1,i} = 6.11 \times 10^6 \Omega$, $R_{2,i} = 0.9 \times 10^6 \Omega$ such that $R_{1,i} = 7R_{2,i}$. Note that we have used $R_{1,i} = R_1$, $R_{2,i} = R_2$, $L_i = L$, and $C_{0,i} = C_0$. For a 1 μm of the AF, we find

$$R_{\text{eff}} = 1.2 \times 10^9 \Omega \quad (18.48)$$

$$L_{\text{eff}} = 340 \times 10^{-12} \text{ } H \quad (18.49)$$

$$C_{\text{eff}} = 0.02 \times 10^{-12} \text{ } F \quad (18.50)$$

This solenoidal flow geometry leads to an equivalent electrical element possessing self-inductance. From Faraday's law, we can derive an effective inductance for the actin filament in solution by

$$L = \frac{\mu N^2 A}{l} \quad (18.51)$$

where l is the length of the F-actin and A is the cross-sectional area of the effective coil given by

$$A = \pi(r_{\text{actin}} + \lambda_B)^2 \quad (18.52)$$

The number of turns is approximated by simply working out how many ions could be lined up along the length of a monomer. We would then be approximating the helical turns as circular rings lined up along the axis of the F-actin. We also take the hydration shell of the ions into account in our calculation. The hydration shell is then the group of water molecules oriented around an ion. It can be shown that $L = 1.7 \text{ } pH$ for the length of the monomer.

The electrical model of the AF is an application of Kirchhoff's laws to one section of the effective electrical circuit that is coupled to neighboring monomers. Taking the continuum limit [148] for a large number of monomers along an AF, we derive the following equation which describes the spatio-temporal

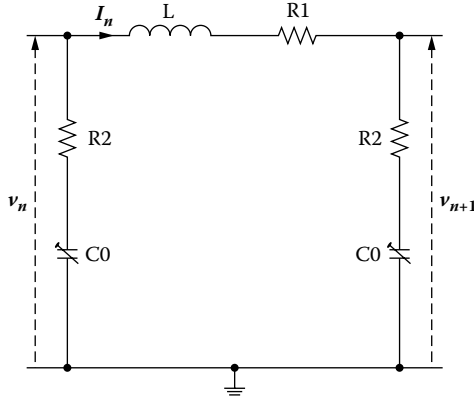


FIGURE 18.17 An equivalent electrical circuit for a segment of an actin filament.

behavior of the potential along the AF:

$$L C_0 \frac{\partial^2 V}{\partial t^2} = a^2 (\partial_{xx} V) + R_2 C_0 \frac{\partial [a^2 (\partial_{xx} V)]}{\partial t} - R_1 C_0 \frac{\partial V}{\partial t} + R_1 C_0 2bV \frac{\partial V}{\partial t} \quad (18.53)$$

Motivated by this picture, physical properties of the ionic distribution along a short stretch of the polymer (the average pitch 35 to 40 nm) have been modeled by Tuszyński *et al.*, [148] as an electrical circuit with non linear components (see Figure 18.17). The main elements of the circuit are: (a) a non linear capacitor associated with the spatial charge distribution between the ions located in the outer and inner regions of the polymer; (b) an inductance due to the helical geometry of the filament; and (c) a resistor due to the viscosity imposed by the solution.

From Kirchhoff's laws, one can derive an equation governing the propagation of voltage along the filament (see [148]). One of the key aspects in this model is the nonlinearity of the associated capacitance [86,150] that eventually gives rise to the self-focusing of the ionic waves. The equations developed for the model originate from the application of Kirchhoff's laws to the RLC resonant circuit of a model actin monomer in a filament. Perhaps the most important finding is the existence of the traveling wave, which describes a moving transition region between a high and low ionic concentration due to the corresponding inter-monomeric voltage gradient. The velocity of propagation was estimated to range between 1 and 100 ms^{-1} depending on the characteristic properties of the electrical circuit model. It is noteworthy that these values overlap with action potential velocities in excitable tissues [55].

Analysis of the model described earlier reveals the possibility of stationary waves in time that may lead to the establishment of spatial periodic patterns of ionic concentration. As mentioned previously, Lader *et al.* [78] applied an input voltage pulse with an amplitude of approximately 200 mV and a duration of 800 μs to an AF, and measured electrical signals at the opposite end of the AF, indicating that AFs support ionic waves in the form of axial nonlinear currents. In an earlier experiment [83], the wave patterns observed in electrically stimulated single AFs were similar to recorded solitary waveforms for electrically stimulated nonlinear transmission lines [85]. Considering the AFs highly nonlinear physical structure and thermal fluctuations of the counter-ionic cloud [105], the observation of soliton-like ionic waves is consistent with the idea of AFs functioning as biological transmission lines.

This section only provides an indication as to a realistic model of actin that can support soliton-like ionic traveling waves. Modeling relies on data constrained by experimental conditions, and/or assumptions made, including the charge density, which is calculated based on the net surface charges of actin. It should also be considered that soliton velocity is directly proportional to the magnitude of the stimulus, which in a biological setting has not been formally described. Actin interacts with a number of ion channels of different ionic permeability and conductance. Thus, it is expected that channel opening, single channel

currents, and other channel properties, including the resting potential of the cell, may significantly modify the amplitude and velocity of the soliton-supported waves. This should correlate with the velocity of the traveling waves along channel-coupled filaments. Other parameters that may play a role in this type of electrodynamic interaction are the local ionic gradients and the regulatory role of actin binding proteins, which can help “focus” the conductive medium, or otherwise impair wave velocity. The physical significance of each of the components of the electrical network (see Figure 18.17) and additional details, are described in [148]. MTs differ from AFs in several respects. One of the key differences is the presence of peptide chains called C-termini that protrude from their surface. In the following section, we outline their role in ionic wave conductivity along MTs.

18.7 Long-Range Spatio-Temporal Ionic Waves Along Microtubules

MTs are long hollow cylinders made of $\alpha\beta$ -tubulin dimers [33], as described in Section 18.5. Recently, it has become apparent that neurons utilize MTs in cognitive processing. Both kinesin and MAP2 that associate with MTs have been implicated in learning and memory [69,154]. Dendritic MTs are implicated in particular, and it is highly probable that the precisely coordinated transport of critical proteins and mRNAs to the post-synaptic density via kinesin along MT tracks in dendrites is necessary for learning, as well as for LTP [70].

The following molecular dynamics (MD) simulation results focus on the C-termini of neighboring tubulins, whose biophysical properties have a significant influence on the transport of material to activated synapses. This affects cytoskeletal signal transduction and processing as well as synaptic functioning related to LTP. Using MD modeling we calculated conformation states of the C-termini protruding from the outer surfaces of MTs and strongly interacting with other proteins, such as MAP2 and kinesin [124]. To elucidate the biophysical properties of C-termini and gain insight into the role they play in the functioning of dendrites, we developed a quantitative computational model based on the currently available biophysical and biochemical data regarding the key macromolecular structures involved, including tubulin, their C-termini, and associated MAP2. In the proposed model of the C-termini microtubular network, the tubulin dimer is considered to be the basic unit. Each dimer is decorated with two C-termini that may either extend outwardly from the surface of the protofilament or bind to it in one of a few possible configurations.

The most dynamic structural elements of the system (i.e., its elastic and electric degrees of freedom) are envisaged as conformational states of the C-termini. Each state of the unbound C-terminus evolves so as to minimize the overall interaction energy of the system. The negatively charged C-termini interact with (a) the dimer’s surface, (b) neighboring C-termini and (c) adjacent MAPs. While the surface of the dimer is highly negatively charged overall, it has positive charge regions that attract the C-termini causing them to bend and bind in a “downward” state. The energy difference between the two major metastable states is relatively small, on the order of a few $k_B T$ [112].

To simplify our calculations, we first used a bead-spring model representing the C-terminus as a sequence of beads with flexible connections. We have accounted for the electric field exerted by the dimer, the external field generated by the environment, and various short-range interactions within the simulated C-terminus, including interactions between the beads (i.e., Lennard–Jones potential, angular forces, etc.).

A simplified model of the interaction between MAP2 and its ionic environment via counter-ions was used to investigate the ability of MAP2 to function as a “wave-guide” that transfers the conformational change in a C-terminus state to an adjacent MT (see Figure 18.18). A perturbation applied to the counter-ions at one end of the MAP2 drives them out of equilibrium and initiates a wave that travels along the MAP2 (for details see [112]). Figure 18.18 (right) depicts the main result for a localized perturbation applied for a few picoseconds to the counter-ions near the binding site of MAP2. Wave propagation along the chain of $N = 50$ counter-ions is shown as a counter-ion displacement parallel to MAP2, u_i , where i denotes the i^{th} counter-ion. This perturbation along MAP2 propagates almost as a “kink” whose phase velocity has been found to be $v_{ph} \approx 2nm \cdot ps^{-1}$.

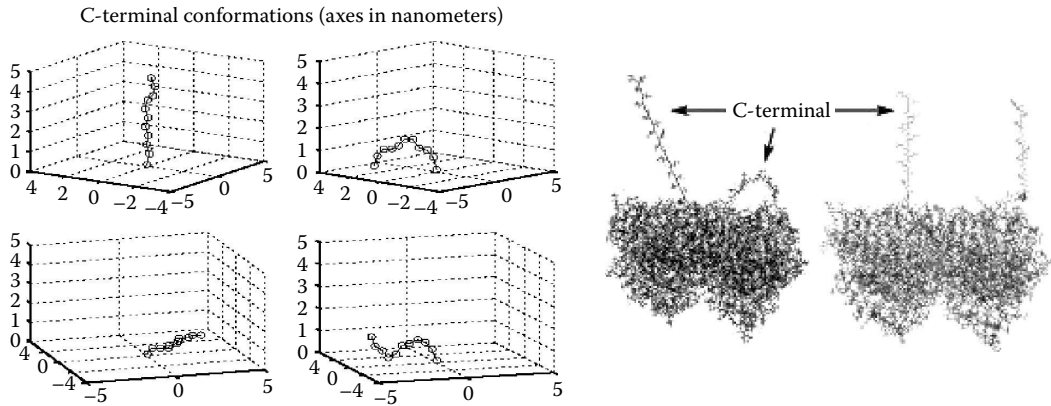


FIGURE 18.18 Examples of the conformational states of the C-termini in a tubulin dimer (right) and results from bead-spring model simulations (left).

Our simulations indicate the ability of an ionic wave to trigger a coupled wave of C-termini state changes from their upright to downward orientations. Four examples of “up” and “down” states of the C-termini obtained through bead-spring model simulations are shown in Figure 18.18 (left) and results from molecular dynamics simulations for an actual tubulin dimer are shown in Figure 18.18 (right). Calculations of the energy minimized positions of the individual beads representing the amino acids of the C-terminus in two equivalent forms reveal that the probability of the down position, which includes all cases of full or partial attachment, is 15%. This means that the system has two major states with a strong bias towards the stretched-up state.

In the remainder of this section, we present an MD simulation study of the interaction between MTs and MAP2, followed by preliminary experimental results of ionic wave conduction by MTs.

A simplified model of the interaction between MAP2 and its ionic environment via counter-ions was used to investigate the ability of MAP2 to function as a “biological wire” that transfers the conformational change in a C-terminus state to an adjacent MT (see Figure 18.19).

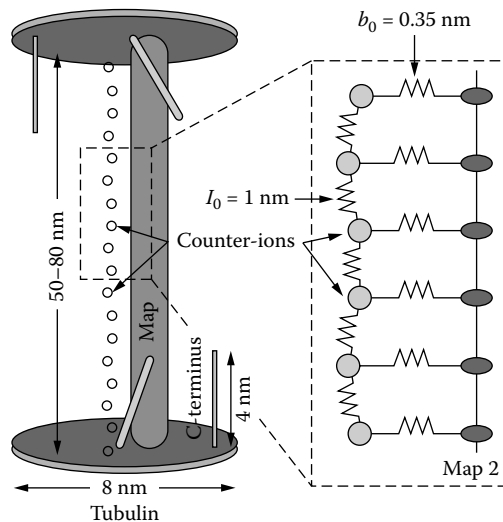


FIGURE 18.19 An illustration of a MT C-terminus conformational change transfer process. Ionic waves travel along an adjacent MAP via local perturbations to the bounded counter-ions.

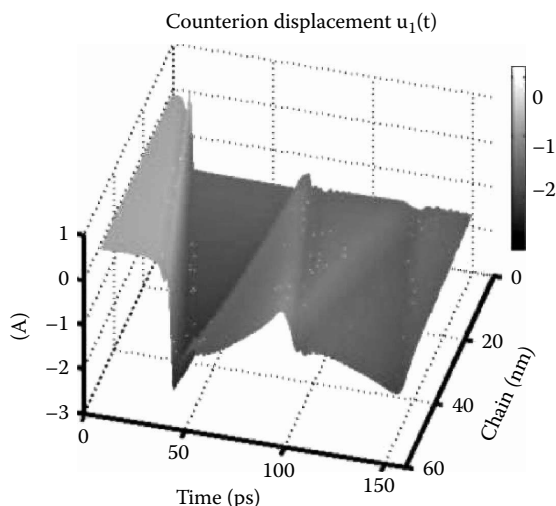


FIGURE 18.20 Propagation of a localized perturbation applied to counter-ions near the binding site of a MAP2.

A perturbation applied to the counter-ions at one end of the MAP2 drives them out of equilibrium and initiates a wave that travels along the MAP2 [112]. Figure 18.20 depicts the main result for a localized perturbation applied for a few picoseconds to the counter-ions near the binding site of a MAP2. Wave propagation along the chain of counter-ions is demonstrated as counter-ions displacement parallel to MAP2.

To validate the proposed model, we have conducted experiments on electro-stimulation of MTs in a solution. A detailed description of the experiment and its results has been published elsewhere [113]. Here we only describe the main aspects of the work on this phenomenon. Figure 18.21 depicts the experimental setup used. Isolated taxol-stabilized MTs have been shown to be able to amplify an electric signal applied to them through a micropipette. The input signal was of 5 to 10 msec duration, with amplitude in the range of ± 200 mV. The signal that arrived at the other end of the MT was more than twice as high as the signal recorded in a control experiment where the same two pipettes were immersed in a solution with no MT making contact to them. The calculated conductivity of MTs was found to be on the order of 10 nS, indicating a high level of ionic conductivity along an MT. By comparison, for a typical ion channel, the corresponding value ranges between 5 and 200 pS.

The results of the MD modeling previously described raise the possibility of transmitting electrostatic perturbations collectively among neighboring C-termini, and from C-termini on one MT to those C-termini on another MT via MAP2. The importance of collective conformational states of C-termini and of transmitting perturbations among the C-termini as a novel information processing mechanism operating at a subneuronal level is clear. MAP2 and kinesin bind near to the C-termini on tubulin and the electrostatic properties of C-termini affect this binding [2,145]. Hence the conformational states of the C-termini must

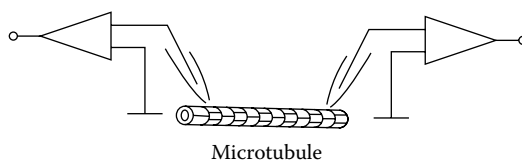


FIGURE 18.21 Schematic representation of the experimental set-up used for the measurements of a MTs ionic conductivity. The MT is attached to two micropipettes that are connected to signal amplifiers.

at some point be taken into account in order to understand neural processing that depends on transport of synaptic proteins inside of neurons. Kim and Lisman [71] have shown that inhibition of MT motor proteins reduces an AMPA receptor-mediated response in hippocampal slice. This means that a labile pool of AMPA-receptors depends on MT dynamics, and MT-bound motors determine the amplitudes of excitatory postsynaptic currents (EPSC). A walking kinesin carries with it a protein or an mRNA molecule. Since kinesin binds to a MT on a C-terminus, as it steps on it, it brings the C-terminus to the MT surface and makes it ineffective in binding for the next kinesin over a period of time that it takes the C-terminus to unbind and protrude outside. From this we can deduce that long stretches of C-termini in the upright position are going to be most efficient at transporting kinesin and kinesin cargo while C-termini that lie flat are expected to be most efficient at detaching kinesin. Thus, in considering the trafficking of many kinesins, collective electrostatic effects of C-termini become crucially important. Collective states that correspond to transport strategies that will send optimal numbers of kinesin molecules to synaptic zones are likely to occur when synapses are activated (e.g., when synapses are generating EPSCs). One type of kinesin cargo associated with learning and memory is the NMDA receptor, which is well known to be associated with LTP. The PSD might be expected to require replenishment of NMDA receptors at a later time than AMPA receptors are replenished (see [71]). Kinesin protein (specifically KIF17) actively transports NMDA receptor 2B subunits (NR2B) to the region of the PSD [52]. Once NR2B is in the vicinity of an active synapse it dissociates from kinesin and then becomes associated with the PSD, presumably using actin transport as an intermediary step. Hence, kinesin-mediated transport of NMDA receptors along MTs has a built-in negative feedback mechanism: whenever too many NMDA receptors are transported to the synaptic site (often located on a spine head), then those NMDA receptors can initiate proteolytic breakdown of MAP2. The latter event would be expected to reduce further transport of NMDA receptors to the synaptic site. The signal transduction molecule CaMKII is also critical for learning and LTP. Similar to the NMDA receptor, CaMKII is transported to active synapses via a kinesin-mediated transport mechanism. However, it appears to be the mRNA for CaMKII to a larger extent than the protein that is transported to spines, since dendrites are enriched with polyribosomes and CaMKII mRNA [140]. Ribosomes are redistributed from dendrites to spines with LTP [106]. Local translation of CaMKII α in dendrites appears to be necessary for the late phase of LTP: fear conditioning and spatial memory [98]. Moreover, changes in synaptic efficacy are often accompanied by changes in morphology; reorganization of underlying MTs is a fundamental factors for these morphological changes. Having discussed the individual properties and roles played by AFs and MTs in subcellular conduction processes, we now turn to an integrated view of their functioning within a cell. The next several sections discuss this issue in depth.

18.8 Dendritic Cytoskeleton Information Processing Model

Our current hypothesis states that the cytoskeletal biopolymers constitute the backbone for ionic wave propagation that interacts with, and regulates, dendritic membrane components, such as ion channels to effectively control synaptic connections. Figure 18.22 depicts a portion of the dendritic shaft where MTs are decorated by C-termini and interconnected by MAP2 (thick line). Connections between MTs and AFs are shown as well as two types of synaptic bindings. On the upper left side, actin bundles bind to the post-synaptic density (PSD) of a spineless synapse. On the lower right side, a spiny synapse is shown, where actin bundles enter the spine neck and bind to the PSD, which at the other end, is connected to the MTN.

We envision a mechanism in which a direct regulation of ion channels and thus synaptic strength by AFs and associated cytoskeletal structures controls and modifies the electrical response of the neuron. In this picture, MTs arranged in networks of mixed polarity receive signals in the form of electric perturbations, from synapses via AFs connected to MTs by MAP2 [122], or via direct MT connections to postsynaptic density proteins by molecules such as CRIPT [108]. As discussed next, the MTN may be viewed as a high-dimensional dynamic system where the main degrees of freedom are related to the conformational state of the C-termini. The input signals perturb the current state of the system that continues to evolve. Hypothetical integration of the preceding ideas is outlined as follows. Electrical signals arrive at the PSD

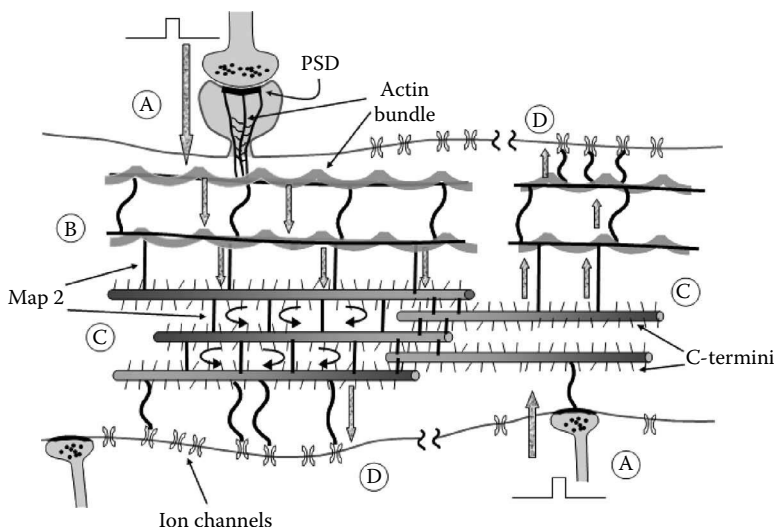


FIGURE 18.22 A scheme of the dendritic shaft with MTs arranged in networks of mixed polarity that receive signals in the form of electric perturbations from synapses via AFs connected to MTs by MAP2, or via direct MT connections to postsynaptic density proteins by molecules such as CRIP1. A spiny synapse is depicted in the bottom right where an actin bundle enters the spine neck.

via synaptic transmission, which in turn elicits ion waves along the associated AFs at the synaptic spine. These dendritic input signals propagate in the form of ionic waves through AFs to the MTN where they serve as input signals. The MTN, operating as a large high-dimensional state machine, evolves these input states, e.g., by dynamically changing C-termini conformation. The output from the MTN is the state of the system that may propagate via AFs to remote ion-channels. These output functions are assumed to regulate the temporal gating state of voltage-sensitive channels. This process subsequently regulates the membrane conductive properties and controls the axon hillock behavior by changing the rate, distribution, and topology of open/close channels. The overall functions of the dendrite and neuron can thus be regulated in this manner.

The attractiveness of the concept that the cytosol, with its cytoskeletal structures, may behave as a large dynamical system is clear since it provides a means for real-time computation without the need for stable attractor states. Moreover, the output is relatively insensitive to small variations in either the MTN (cytoskeletal networks) or the input patterns. It should be noted that the temporal system's state evolves continuously, even without external inputs. Recent perturbations, however, have a long-term effect on the MTN trajectories, i.e., there is a memory effect inherent to this system (not to be mistaken with synaptic LTP, which has a much longer time scale). The output from the MTN may converge at or near ion channels to regulate their temporal behavior. The issue of adaptation requires a feedback mechanism that will, at least locally, enable the change of the output function. In the context of neuronal function, with a focus on processivity, synaptic strengthening, LTP, and memory enhancement, the output function may simply reflect an effect of the MTN on synaptic channel function, such that the desired state of the channel appears to have a higher probability of being open/close upon the presentation of the associated input pattern. One possibility is a Hebbian-based response where more frequent activity of certain subdomains of the MTN output states gives rise to a higher/lower density of AFs connecting to corresponding channels.

We now explain how this integrative view may serve as a regulatory (adaptive) mechanism. The input level denoted by A in Figure 18.22 is associated with external electric perturbations passing through membranes, mainly synaptic inputs arriving from other neurons. These signals arrive at the cytoskeleton and directly affect MTNs, and/or actin filaments (bundles), in spiny synapses. However, the actin cytoskeleton is responsible for the propagation of signals to the MTN (see level B). These propagated signals (level C)

are in turn used as inputs to the MTN, viewed as a dynamic system. We further propose that the MTN generates diverse phase space trajectories in response to different input vectors. The requirements from such a system are not too restrictive since the output-state is not an attractor of the system. In other words, information processing at this level is not necessarily based on attractor dynamics but rather on real-time computations. This proposition relies on the observed ability of the MTN to propagate signals and on the specific topological features of MTNs in dendrites, in which shorter MTs of mixed polarity are interconnected by MAP2s. The output from the MTN would be a function of the evolved state vector in certain areas accessed by actin filaments and/or directly linked to ion channels (see both possibilities at level D). These output signals may modify the temporal channel activity, either by directly arriving from the MTN to the channels, or mediated by actin filaments. Hence, the channel-based synaptic membrane conductance is regulated, in particular in the axon hillock region, which is, in most cases, responsible for the generation of action potentials.

The idea that a nonspecific high-dimensional dynamical system may serve as a reservoir of trajectories in the context of liquid state machines (LSMs), has recently been suggested as an explanation for the existence of microcircuits in the brain [87]. The basic structure of an LSM is an excitable medium (hence “liquid”) and an output function that maps the current liquid state. The liquid must be sufficiently complex and dynamic to guarantee a universal computational power. This is sufficient to ensure that different input vectors lead to separate trajectories. A network of spiking neurons and a recurrent neural network has been used as a “liquid”, whereas the output (readout) function has been implemented by simple perceptrons, threshold functions, or even linear regression functions. Clearly, simpler readout functions restrict the ability of the whole system to capture complex nonlinear dependencies.

18.9 The Inter-Relation Between the Neural Cytoskeleton and the Membrane

Usually, F-actin and microtubular cytoskeletal networks are thought to fulfill separate, independent cellular roles. Highly dynamic actin networks are known for their role in cell motility, in particular the spreading of the leading edge and contraction. The more stable MT cytoskeleton is best known for its importance in cell division and organelle trafficking. However, recent studies provide a more unified role, ascribing important roles to the actin cytoskeleton in cell division and trafficking and important roles for MTs in the generation and plasticity of cellular morphology. Coordination between the actin- and MT- based cytoskeletons has been recently observed during cellular migration and morphogenesis, processes that share some similarities with neurite initiation [28]. A direct physical association between both cytoskeletons has been suggested, because MTs often preferentially grow along actin bundles and transiently target actin-rich adhesion complexes. In neurons, certain plakins and neuron-specific MAPs, like MAP1B and MAP2, may play a role in linking MTs and AFs, helping in the transition from an undifferentiated state to neurite-bearing morphology.

MAP1B and MAP2 are both known to interact with actin *in vitro* [20,109,128,134,146]. It is likely that by crosslinking, MAP2 and/or MAP1B associated with both cytoskeletons could be involved in guidance of MTs along AF bundles. Alternatively, MAPs could shuttle from MTs to actin and could alter F-actin behavior by actively crosslinking AFs.

Little is known, however, about interactions between neuronal ion channels and the cytoskeleton. Recently, whole-cell and single channel recordings showed that acute disruption of endogenous AFs with cytochalasin D activated voltage-gated K^+ currents in these cells, which was largely prevented by intracellular perfusion with the AF-stabilizer agent, phalloidin. Interestingly, the direct addition of actin to excised, inside-out patches activated, and/or increased single K^+ channels. Thus, acute changes in actin-based cytoskeleton dynamics regulate voltage-gated ion channel activity in bipolar neurons. This may be indicative of a more general and quite appealing mechanism by which cytoskeletal structures control feedback mechanisms in neuronal channels.

Recent theoretical and experimental studies of the electrical properties of AFs and MTs in solution revealed their capability to act as biopolymer wires [83,148]. This means that these protein polymer

filaments are capable of conducting nonlinear ionic waves and even amplifying the signal with respect to the conducting solution [112]. We conjecture a mechanism in which a direct regulation of ion channels and thus synaptic strength by AFs and associated cytoskeletal structures controls and modifies the electrical response of the neuron. According to this scenario (see Figure 18.22), MTs arranged in networks of mixed polarity receive signals in the form of electric perturbations from synapses via AFs connected to MTs by MAP2 [122], or via direct MT connections to postsynaptic density proteins by molecules such as CRIPT [108]. These signals propagate in the form of ionic waves. Specific physical properties of the propagating ionic waves will be discussed in sections that follow.

18.10 Relationship to Cognitive Functions

Transport of proteins and receptors in neurons is likely to have an electromagnetic basis to the extent that this function is possibly a result of MT computation. The perspective advanced in this chapter is that a specific fingerprint defined by a particular electromagnetic state of a microtubular array potentially corresponds to a unique unit of cognition (e.g., a basic visual parameter). Recently, it has been shown that visual components can be represented musically [23]; hence, the idea that one common type of energy underlies divergent perceptual and cognitive processes appears likely. Activation of one electromagnetic fingerprint could, in turn, activate another electromagnetic fingerprint, irrespective of sensory input. Moreover, the subjective feels of this widespread pattern of electromagnetic energy can be specified according to those key physical properties of MTs that influence the transport of proteins to synapses. Factors influencing kinesin-mediated transport include the protein conformation of tubulin and the nature of the C-termini (see [112,136]).

Not only is the conformational state of tubulin critical to effective transport, motor proteins appear to alter albeit temporarily, the conformation of tubulin. Kinesin binding and that of MAP, tau, significantly alter the direction of the protruding protofilament ridges along MTs, which in turn influences their further binding abilities [127]. More than mere local adaptation to binding, MTs may alter their conformation ahead of kinesin processivity [76], supporting the notion of long-range cooperative effects between tubulin dimers located along longitudinal protofilaments of MTs. These biochemical relationships have consequences for electromagnetic fields. The dipole moment of tubulin depends on its configuration in the MT [53,97]. Thus, electromagnetic fields among MTs could, in theory, be induced or inhibited by synaptic inputs that affect the protein conformation of tubulin directly or through alterations in kinesin or MAP binding. Synaptic effects upon MTs could be mediated through ionic currents, by propagation via actin filaments [148], or by signal transduction cascades resulting in the phosphorylation of MAPs [126].

Due to lengthwise electric dipoles of tubulin dimers, information in the form of traveling waves propagated along microtubular tracks can, in principle, be transmitted between synapses with high fidelity (see [149]). MAP2 bridges keep microtubular arrays within the dendritic core parallel and antiparallel by aligning portions of polarized MTs. The antiparallel alignment of MTs, which specifically occurs in dendrites, would severely attenuate any electromagnetic field generated by MTs. However, during enhanced kinesin-mediated transport, as is likely to occur with heightened synaptic activity, MAP2 bound to the MT would be perturbed and may even temporarily detach from the MT. A similar phenomenon might also occur due to dynein-mediated transport, which occurs largely in the opposite direction to that of kinesin-mediated transport. Assuming that at least some MAP2 stays attached to the antiparallel MTs, keeping the dendritic array intact, any net unidirectional transport along the MT array should increase the strength of the electromagnetic field associated with the fingerprint and should further result in the spread of that electromagnetic field to adjacent MTs. Once a sufficient number of MTs were engaged in dynamically sending and receiving complementary electromagnetic energies, whole neuronal compartments (e.g., dendrites) might be expected to interact. Due to the parallel/antiparallel arrangement of MTs in cortical dendrites and the ability of electromagnetic fields to pass from one dendrite to adjacent dendrites, information could, in principle, pass between neurons when such electromagnetic fields were sufficiently amplified as a result of changes in the binding of MAPs or kinesin.

While it is true that there are some 100 billion neurons and some 100 trillion synapses in the human cerebral cortex, so too does each sensory field afford the luxury of extremely high resolution. The fovea of the retina, for example, contains roughly 1 million receptor cells that relay information to the cerebral cortex, diverging to eventually drive the synaptic activity of at least 1 billion cortical cells. Although the mammalian cortex has many neurons, and even more synapses, its connectivity can be readily grasped by one recurrent theme: repetitive high-resolution topographic representation. Despite this simple organizational plan, cells in cortical areas, in particular cells in higher sensory or association areas, inexplicably show correlated responses during cognitive tasks. Experience alters the basic structure of dendrites, and as a consequence, it should also alter any electromagnetic fields generated by MTs as a result of transport proteins moving along them. As described earlier, a rearrangement of the cytoskeleton occurs during early development, with learning, and with neurodegeneration underlying dementia.

Although models of changes in synaptic efficacy, e.g., long-term potentiation or depression (LTP or LTD), offer great potential as memory mechanisms (see [93]), one often overlooked problem is that large numbers of synapses are affected in concert. If only a few synapses are changed with each memory, then the entire cortical system would have a near unlimited capacity, but if many synapses participate in each memory, as seems to be the case for LTP, then there could be a serious saturation effect. This relates back to the topographic organization of the cerebral cortex. Strictly synaptic models (electrochemical) suggest that complex neural networks increase synaptic weights (i.e., neurons that fire together wire together). Nonetheless, visual cortical regions, for example, do not have massively random interconnections among all parts of the visuotopic map, thereby making many of the changes in synaptic efficacy necessary for encoding complex perceptual features impossible.

The biophysical properties of MTs are just beginning to be understood at molecular and atomic levels, and recent empirical evidence suggests interesting effects occur between MTs under certain experimentally induced conditions. Two groups, one led by Watt Webb at Cornell and another led by Paul Campagnola and William Mohler at the University of Connecticut, observed that MTs give rise to intense second-harmonic generation: a frequency doubling upon exposure to a sapphire laser in the 880-nm range. (Other frequencies were partially effective.) MTs were one of the few biological materials having electric dipoles that constructively interfered with the dipoles of neighboring MTs [11,29]. This occurred for parallel MTs in axons, but not for antiparallel MTs in dendrites. Could such a phenomenon be expected to occur with natural learning or upon exposure to oscillatory input, an LTP-inducing tetanus or various pharmacological agents? To the extent that these induce electromagnetic energy, it is conceivable. In addition to being sensitive to electromagnetic radiation, MTs may themselves produce this kind of energy. Second-harmonic generation by MTs is consistent with their ferroelectric properties. In second-harmonic generation, the most strongly enhanced wavelength is 880-nm, supporting the frequently overlooked reports of electromagnetic signaling by cells. Guenther Albrecht-Buehler [5] observed electromagnetic energy in the near infrared region overlapping the 880 nm value that was generated by centrioles for the purpose of cell-to-cell communication, leading him to suggest that: "... one of the functions of MTs may be to play the role of cellular 'nerves.'" Infrared light in this range of wavelengths also induces cell aggregation [6]. Since frequency (and hence wavelength) determines the long-range effect of dielectric polarization of a given MT, one would expect this effect to be length-dependent. The distance between MTs was shown to be critical in second-harmonic generation by MTs.

18.11 The Potential for Bioelectronic Applications and Neuromorphic Computing

Traditionally, biologists viewed the cell as a "fluid mosaic bag" containing enzymes, organelles, etc. Recent work has focused on finding the proteins that come together in this cellular "soup" and proceed to effect a change corresponding to a particular biological function. Through the molecular-level analysis of these protein-protein interactions, recent work in various labs has proceeded to build a model of "systems biology," in which these biochemical reactions that have been individually characterized can be brought together to form a cohesive whole, a synergistic dynamics network. From this knowledge, we hope to extract

useful information that will affect fields as diverse as nanotechnology, biotechnology, and materials science. Inside each cell, there is an elaborate network that serves a multitude of functions. It is an intracellular “highway” that provides a roadway for motor proteins to carry cargo and travel along in their journey to extend axonal length, for example. It controls synapses, and dendrite outgrowth. It is responsible for pulling the chromosomes apart during mitosis, for cell motility, etc.

As man-made engineered systems become more and more miniaturized, the push to build “nanomachinery” is driving physicists and engineers towards biomimetics, (i.e., the branch of engineering that mimics biological materials). This is highly feasible and much progress has been made in this area. However, due to the limitations inherent in trying to build mechanical devices on such a small scale, there is still much to learn. It is conceivable that through our better understanding of the cell’s individual functional and structural components we will be able to advance medicine and technology beyond even our wildest dreams. One area that may undergo a major paradigm shift as a result of it is computer engineering.

As we enter into the 21st century, the dawn of the exciting new world of computational biotechnology is upon us and we are poised for a new generation of computers that do not have to be told what to do, but have the power to learn on their own. The development of a biological computer that is fast, small, and evolvable is no longer considered science fiction. Scientists are combining biological materials with the latest silicon-based technology in an effort to give us not only electronic devices that are smaller, but are also more flexible in terms of structure and function.

Ironically, our brain can serve as proof of concept. Many scientists now believe that individual neurons display wiring patterns and communication powers that resemble a human computer, and therefore may provide the framework for the design of a biological computer. The brain is composed of 10 billion neurons, each of which may be communicating with 1000 others. The neuron transmits electric signals along its arms, or ‘axons’. Inside each axon a parallel architecture of microtubules is interconnected with other proteins, not unlike the parallel computer’s wiring. In fact, the structure of microtubules may have evolved towards optimal computational efficiency. The piezoelectric properties of these protein filaments allow them to bend as a result of electric fields or currents which makes them ideal candidates as regulators of synaptic plasticity, a mechanism that could explain the popular principle of “use it or lose it.” Thus, seldom used synaptic connections would be switched off in favor of more active ones. Biologist Guenter Albrecht-Buehler [3] demonstrated that cells perceive their environment through a tiny organelle called the centriole, composed of microtubules that were shown to interact with electromagnetic radiation. To further display that microtubules themselves can be conductive, the German biotechnology group, led by Eberhard Unger [41,42] successfully conducted required experiments and is now working on building nanoelectronic components using these and other proteins. Simultaneously, ideas of combining biological and silicon-based materials in hybrid arrangements are gaining more and more support for future bioelectronic applications. Work is underway at various labs around the world to develop a biological chip. The main objective is to design nanoelectronic components using hybrid protein–silicon structures with arrays of proteins as biological oscillators that can be stimulated by electrodes or acoustic couplers. This will provide an analog circuit with rich dynamics.

DNA computers are unlikely to become stand alone competitors for electronic computers. But digital memory in the form of DNA and proteins is a real possibility with exquisitely efficient editing machines that navigate through the cell, cutting and pasting molecular data into the stuff of life. Beyond that, the innate intelligence built into DNA molecules could help fabricate tiny, complex structures: in essence, using computer logic not to crunch numbers but to build things, an idea conceived of by Caltech’s Winfree and Rothmund [123]. A single test tube of DNA tiles could perform about 10 trillion additions per second (about a million times faster than an electronic computer). Interfacing DNA or protein-based hybrid structures with living cells will enable a host of medical and technological applications including non-invasive diagnostic and therapeutic medical applications and computational devices.

One major component of this framework is a protein-based subcellular structure called a microtubule, to which we have devoted a large portion of this chapter. Microtubules are composed of 13 protofilaments, arranged in a spiral pattern, and each protofilament is composed of subunits of tubulin. Each tubulin possesses one of two conformational states: either both monomers can align, or one monomer can shift

30° relative to the vertical axis. Based on the conformation an individual tubulin exists in at a given moment, the conformation of its nearest neighbors can be altered as well. This conformational change can be transmitted throughout the microtubule in a similar manner as patterns are transmitted through cellular automata programs. In addition, as we argued earlier in this chapter, tubulin can have several electronic states as well as dipolar states, some of which can be conceivably linked to the conformations of C-termini and the counter-ion clouds around them. The idea that we could input an electric, chemical, or mechanical signal and generate another one at the output end of the system brings computing or information processing to mind. We believe that ordered protein arrays can function as a programmable processor or even a computer either inside cells such as neurons or as nano-scale devices.

Nanotechnology could and should take advantage of biology in a completely novel manner, (i.e., instead of building materials that mimic biological components, we could use the components themselves). In the case of microtubules and tubulin, rather than purify it from traditional sources such as cow or pig brains, tubulin can be readily cloned into an expression vector, and purified from bacteria in mass quantities. Then, purified tubulin of a predetermined sequence (and thus with desirable biophysical and biochemical characteristics) can be routinely polymerized to form microtubules with specific structural and functional properties. Following that, the microtubules produced could be stabilized in a polymer gel capable of undergoing a phase transition at a particular value of the control parameter such as temperature, applied pressure, pH, or ionic concentration. This would then trigger a signal that would initiate computational processes carried out by individual microtubules or their assemblies interconnected by MAPs or even connected to actin filaments in architectures that could mimic subneuronal designs. This technology could also be applied to sensors or artificial intelligence systems in building an evolvable computer. The only limitations in this endeavor are those imposed by our imagination and our ability to dream up new applications.

18.12 Discussion

Considering the abundance of MTNs and AFs in axons and dendritic trees, the findings and theoretical models described earlier may have important consequences for our understanding of the signaling and ionic transport at an intracellular level. Extensive new information (see [63]) indicates that AFs are both directly [14] and indirectly linked to ion channels in both excitable and non-excitable tissues, providing a potentially relevant electrical coupling between these current generators (i.e., channels), and intracellular transmission lines (i.e., AFs and MTs). Furthermore, both filaments are crucially involved in cell motility and, in this context, they are known to be able to rearrange their spatial configuration. In nerve cells, AFs are mainly located in the synaptic bouton region, whereas MTs are located in both dendrites and the axon. Again, it would make sense for electrical signals supported by these filaments to help trigger neurotransmitter release through a voltage-modulated membrane deformation leading to exocytosis [132]. Actin is also prominent in post-synaptic dendritic spines, and its dynamics within dendritic spines has been implicated in the post-synaptic response to synaptic transmission. Kaech *et al.*, [65] have shown that general anesthetics inhibit this actin mediated response. Among the functional roles of actin in neurons, we mention in passing glutamate receptor channels, which are implicated in long-term potentiation. It is therefore reasonable to expect ionic wave propagation along AFs and MTs to lead to a broad range of physiological effects.

In summary, this chapter broadly described the physical conditions that enable cytoskeletal polymers such as AFs and MTs to act as electrical transmission lines for ion flows along their lengths and along MAP-mediated interconnections between adjacent filaments. In the case of AFs, we propose a model in which each protein subunit is equivalent to an electric circuit oscillator. The physical parameters used in the model were evaluated based on the molecular properties of the polymer. Using the general conductivity rules that apply to electrical circuits we analyzed the properties of ionic waves that propagate along AFs and compared these values to those observed in earlier experiments. In the context of the role played by MTs in neurons, we described the dynamics of C-termini states. We discussed both individual MTs and their networks, including the interactions with ions and signal transmission via MAPs. Recent experiments

on ionic conductivity along AFs and MTs show the validity of the basic assumptions postulated in our models. In light of these results, we conjectured a new dendritic signaling mechanism that involves ion waves along protein filaments which may travel without significant decay over tens of microns, potentially affecting the function of synapses and ion channels.

We proposed a new model for information processing in dendrites based on electrical signaling involving the cytoskeleton. This model predicts that the dendritic cytoskeleton, including MTs and AFs plays an active role in computations affecting neuronal function. These cytoskeletal filaments are affected by, and regulate, ion channel activity. A molecular dynamics description of the C-termini protruding from the surface of a MT has revealed the existence of several conformational states, which lead to collective dynamical properties of the neuronal cytoskeleton. These collective states of the C-termini on MTs have been shown to have a significant effect on the ionic condensation and ion cloud propagation with physical similarities to those recently found in actin-filaments. We have been able to then provide an integrated view of these phenomena in a bottom-up scheme that demonstrates how ionic wave propagation along cytoskeletal structures may impact channel functions, and thus neuronal computational capabilities.

The possibility of an evolvable, dynamic, and responsive electrical circuitry within the cell provided by actin and MT filaments could be of enormous consequence to our understanding of the way cells operate internally and interact with their environment. In particular, it would cast an entirely new light on cell differentiation, cell division, and cell–cell communication. While an integrated theory of this type of behavior is far from being constructed, its individual elements are gradually taking shape.

Acknowledgments

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